

MOTION IN A STRAIGHT LINE

Student Practice · Concept + Difficulty Edition

Attempt -> work -> optional hints -> concept-assimilating method check

HOW THE BADGES WORK

EASY

MEDIUM

HARD

CHALLENGE

Source D1-D4 remains preserved internally; students see the word badge.

PHASE 1 · ASSIMILATION

Concept is visible. Questions rise in reasoning demand inside each concept family.

PHASE 2 · MIXED TRANSFER

Difficulty remains visible, but the concept is hidden until the method check.

YOUR ROADMAP • ONE SET AT A TIME

Do not browse the whole book. Finish the current set, check methods, then move on.

ASSIMILATION 1/7	Averages & acceleration basics	8 questions
ASSIMILATION 2/7	Constant acceleration core	7 questions
ASSIMILATION 3/7	Polynomial motion, intervals & braking pairs	8 questions
ASSIMILATION 4/7	Free fall & vertical stages	8 questions
ASSIMILATION 5/7	Relative motion & paired launches	4 questions
ASSIMILATION 6/7	Graph reading & graph families	9 questions
ASSIMILATION 7/7	Advanced graph models & interval review	5 questions
MIXED TRANSFER 1/2	Mixed transfer I	6 questions
MIXED TRANSFER 2/2	Mixed transfer II	6 questions

Q1 EX-M2B-02

EASY

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2023 - 10 APR EVENING

A person travels distance x at velocity v_1 and then another equal distance x at velocity v_2 in the same direction. Relate the average velocity v to v_1 and v_2 .

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Both legs have the same distance x . Their times are x/v_1 and x/v_2 .

H2 · MODEL

Total distance = $2x$; total time = $x/v_1 + x/v_2$.

H3 · START

$v_{av} = 2x / (x/v_1 + x/v_2)$. Cancel x and simplify.

EXAMSIDE SOURCE

CHECK METHOD ->

Q2 EX-M2B-04

EASY

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2023 - 30 JAN EVENING

A vehicle travels 4 km at 3 km/h and another 4 km at 5 km/h. Find the average speed.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The two distances are equal: 4 km each. Times are $4/3$ h and $4/5$ h.

H2 · MODEL

Total time = $4/3 + 4/5 = 32/15$ h; total distance = 8 km.

H3 · START

Average speed = $8 / (32/15)$. Now simplify.

EXAMSIDE SOURCE

CHECK METHOD ->

Q3 EX-M2B-05

EASY

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2023 - 25 JAN MORNING

A car travels distance x at speed v_1 and then the same distance x at speed v_2 . Find average speed.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The two legs are equal distances x . Their times are x/v_1 and x/v_2 .

H2 · MODEL

Total distance = $2x$; total time = $x/v_1 + x/v_2$.

H3 · START

Average speed = $2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q4 EX-M2B-03

MEDIUM

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2023 - 1 FEB MORNING

An object covers three equal consecutive distances at speeds v_1 , v_2 and v_3 . Find the average speed.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Let each equal distance be L . The three travel times are L/v_1 , L/v_2 and L/v_3 .

H2 · MODEL

Total distance = $3L$; total time = $L(1/v_1 + 1/v_2 + 1/v_3)$.

H3 · START

$v_{avg} = 3L / [L(1/v_1 + 1/v_2 + 1/v_3)]$. Cancel L .

EXAMSIDE SOURCE

CHECK METHOD ->

Q5 EX-M2B-06

MEDIUM

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2025 - 2 APR MORNING

A person moves distance x at 5 m/s and the next $3x/2$ at speed v_2 . The overall average velocity is $50/7$ m/s. Find v_2 .

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The distances are x and $3x/2$, so total displacement = $5x/2$.

H2 · MODEL

Total time = $x/5 + 3x/(2v_2)$.

H3 · START

Set $(5x/2)/(x/5 + 3x/(2v_2)) = 50/7$ and solve for v_2 .

EXAMSIDE SOURCE

CHECK METHOD ->

Q6 EX-M2B-08

MEDIUM

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2022 - 28 JUN EVENING

A car covers three equal one-third portions at 11, 22 and 33 m/s. Find average velocity.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The three portions are equal distances. Use one common distance L .

H2 · MODEL

Total time = $L/11 + L/22 + L/33$; total distance = $3L$.

H3 · START

Average = $3/(1/11 + 1/22 + 1/33)$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q7 EX-M2B-01

HARD

M-ST02B · Equal Time vs Equal Distance Averages

JEE MAIN 2024 - 9 APR MORNING

Half the journey is at 6 m/s. The remaining half is covered in two equal time intervals at 9 m/s and 15 m/s. Find the average speed.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The second half uses equal times at 9 and 15 m/s, so its speed for that half is $(9 + 15)/2 = 12$ m/s.

H2 · MODEL

Let total journey = $2L$. First-half time = $L/6$; second-half time = $L/12$.

H3 · START

Average speed = $2L / (L/6 + L/12)$. Now simplify.

EXAMSIDE SOURCE

CHECK METHOD ->

Q8 EX-M3A-01

HARD

M-ST03A · Acceleration as Change in Velocity

JEE MAIN 2023 - 29 JAN MORNING

A tennis ball is dropped from 9.8 m and rebounds to 5.0 m. Contact with the floor lasts 0.2 s. Take $g=10$ m/s². Find average acceleration during contact.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Choose upward positive. Just before impact the ball is moving downward; just after rebound it is moving upward.

H2 · MODEL

From the given heights, speeds are 14 m/s before impact and 10 m/s after rebound, so $v_{\text{before}} = -14$ and $v_{\text{after}} = +10$ m/s.

H3 · START

Average acceleration = $\Delta v / \Delta t = [10 - (-14)] / 0.2$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q9 EX-M4A-01

EASY

M-ST04A · Constant Acceleration and $v = u + at$

TRANSFER QUESTION JEE Main 2023, 6 Apr Evening

A particle moves in +x with $u = 10$ m/s and constant $a = 2$ m/s². How long to reach 60 m/s?

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Distance is irrelevant. The known quantities are $u=10$ m/s, $v=60$ m/s and $a=2$ m/s².

H2 · MODEL

Use the constant-acceleration relation $v=u+at$.

H3 · START

Substitute $60 = 10 + 2t$ and solve for t .

EXAMSIDE SOURCE

CHECK METHOD ->

Q10 EX-M4B-01

MEDIUM

M-ST04B · Displacement under Constant Acceleration

TRANSFER QUESTION JEE Main 2024, 4 Apr Evening

A bus moving at 72 km/h is brought to rest uniformly in 4 s. Find the braking distance.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

72 km/h = 20 m/s. "Brought to rest" gives $v = 0$.

H2 · MODEL

One uniform-braking interval. First find a from $v = u + at$, then find s .

H3 · START

Convert 72 km/h = 20 m/s. With $v = 0$ and $t = 4$ s, $0 = 20 + 4a \rightarrow a = -5$ m/s². Then $s = ut + \frac{1}{2}at^2 = 20(4) + \frac{1}{2}(-5)(16) = 80 - 40 = 40$ m.

EXAMSIDE SOURCE

CHECK METHOD ->

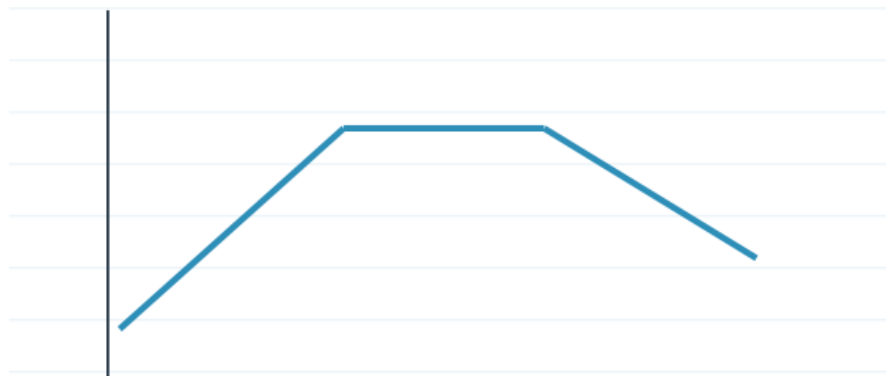
Q11 EX-M4B-04

MEDIUM

M-ST04B · Displacement under Constant Acceleration

TRANSFER QUESTION JEE Main 2019, 10 Jan Evening

A particle starts at the origin and moves along +x with the shown v-t graph. Find its position at $t = 5$ s.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Position change equals signed area under v-t.

H2 · MODEL

Break the graph into simple rectangles/triangles and add their signed areas.

H3 · START

The particle starts from $x = 0$, so its position after 5 s equals the signed area under the given v-t graph from 0 to 5 s. Break the graph into its simple pieces and add their areas. ExamSIDE gives the resulting position as 9 m.

EXAMSIDE SOURCE

CHECK METHOD ->

Q12 EX-M4C-01

MEDIUM

M-ST04C · Average Velocity under Constant Acceleration

TRAIN: ACCELERATE, THEN CRUISE JEE Main 2024, 6 Apr Morning - adapted

A train starts from rest, accelerates uniformly to 80 km/h during time t , then continues at 80 km/h for another $3t$. Find the average speed for the whole $4t$ journey.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The first stage has uniform acceleration, so its average speed is the midpoint of 0 and 80.

H2 · MODEL

Stage 1 distance = $40t$. Stage 2 distance = $80(3t)$.

H3 · START

Stage 1 is uniform acceleration from 0 to 80, so $v_{avg,1} = 40$ km/h. Hence $s_1 = 40t$. Stage 2 is constant 80 km/h for $3t$, so $s_2 = 240t$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q13 EX-M4D-01

MEDIUM

M-ST04D · Remove Time: Why Velocities Become Squared

STOPPING DISTANCE WHEN SPEED DOUBLES JEE Main 2018, 15 Apr Morning - adapted

An automobile travelling at 40 km/h can be stopped in 40 m under a fixed braking condition. Under the same condition, what stopping distance is needed at 80 km/h?

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The braking condition is unchanged, so $|a|$ is the same.

H2 · MODEL

Speed doubles: $u_2/u_1 = 2$. Stopping-distance ratio is the square of this speed ratio.

H3 · START

For fixed braking acceleration, stopping distance $\propto u^2$. The speed ratio is $80/40 = 2$, so the stopping-distance ratio is $2^2 = 4$. Therefore the required distance is $4 \times 40 = 160$ m.

EXAMSIDE SOURCE

CHECK METHOD ->

Q14 EX-M4D-02

HARD

M-ST04D · Remove Time: Why Velocities Become Squared

THE TRAIN BRAKES TOO LATE JEE Main 2023, 1 Feb Evening - adapted

A train moving at 20 m/s normally needs 500 m to stop under a fixed retardation. If braking begins only 250 m before the station, the train crosses the station at speed (x) m/s. Find x.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The first stopping case determines the constant retardation.

H2 · MODEL

Use $v^2 = u^2 + 2as$ on the 500 m stop to get a.

H3 · START

Reuse the same a over 250 m, then identify x from v^2 .

EXAMSIDE SOURCE

CHECK METHOD ->

Q15 EX-M4D-03

HARD

M-ST04D · Remove Time: Why Velocities Become Squared

BULLET PENETRATION UNDER CONSTANT RESISTANCE JEE Main 2022, 27 Jul Evening - adapted

A bullet's speed becomes one-third of its entry speed after penetrating 4 cm into a wooden block. Assume constant resistance. If it finally stops after travelling (4 + x) cm in the block, find x.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Constant resistance -> constant retardation. Distance, not time, is the natural variable.

H2 · MODEL

Compare the loss in v^2 over the first 4 cm with the remaining loss from $(u/3)^2$ to 0.

H3 · START

First loss = $u^2 - u^2/9 = 8u^2/9$ over 4 cm.

EXAMSIDE SOURCE

CHECK METHOD ->

Q16 EX-M4E-01

MEDIUM

M-ST04E · Read a Motion Equation from Its Coefficients

JEE MAIN 2023 - 15 APR MORNING

$x(t)=5t^2-4t+5$ m. Find the magnitude of velocity at $t=2$ s without calculus.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Compare $x(t)=5t^2-4t+5$ with $x=x_0+ut+(\frac{1}{2})at^2$.

H2 · MODEL

Read $x_0=5$, $u=-4$ and $a=10$ m/s².

H3 · START

At $t=2$ s, use $v=u+at = -4 + 10(2)$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q17 EX-M4E-02

MEDIUM

M-ST04E · Read a Motion Equation from Its Coefficients

JEE MAIN 2023 - 13 APR EVENING

$s(t)=2.5t^2$. Find instantaneous speed at $t=5$ s using the constant-acceleration form.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Compare $s(t)=2.5t^2$ with $x=x_0+ut+(\frac{1}{2})at^2$.

H2 · MODEL

$u=0$ and $(\frac{1}{2})a=2.5$, so $a=5$ m/s².

H3 · START

At $t=5$ s, $v=u+at = 0 + 5(5)$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q18 EX-M4E-03

MEDIUM

M-ST04E · Read a Motion Equation from Its Coefficients

JEE MAIN 2023 - 25 JAN EVENING

Distance from the start is $x=4t^2$. Find velocity at $t=5$ s under the constant-acceleration interpretation.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Compare $x=4t^2$ with $x=x_0+ut+(\frac{1}{2})at^2$.

H2 · MODEL

$u=0$ and $(\frac{1}{2})a=4$, so $a=8$ m/s².

H3 · START

At $t=5$ s, $v=u+at = 8(5)$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q19 EX-M5A-02

MEDIUM

M-ST05A · Distance During the nth Second

THE NEXT t SECONDS JEE Main 2022, 29 Jun Evening - adapted

A small toy starts from rest under constant acceleration. It travels 10 m in the first t seconds. How far does it travel in the next t seconds?

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

"Next t seconds" means the second interval only: $S(2t)-S(t)$.

H2 · MODEL

From rest with constant acceleration, $S(t)$ is proportional to t^2 . If $S(t)=10$, then $S(2t)=4S(t)=40$.

H3 · START

Distance in the next t seconds = $40 - 10$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q20 EX-M5A-01

HARD

M-ST05A · Distance During the nth Second

TWO nth-SECOND DISTANCES JEE Main 2024, 4 Apr Morning - adapted

A body travels 102.5 m during its nth second and 115.0 m during its (n+2)th second. Find its constant acceleration.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Both data are interval displacements, not cumulative totals.

H2 · MODEL

Use $s_n = u + (a/2)(2n-1)$ for both intervals.

H3 · START

Subtract: $115.0 - 102.5 = s_{n+2} - s_n = 2a$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q21 EX-M5B-01

HARD

M-ST05B · Why Successive Distances Follow 1 : 3 : 5 : 7

SUBTOPIC COVERAGE AUDIT Primary ownership stays unique AUDIT Primary required question

A body starts from rest with constant acceleration. It covers S_1 in the first (p-1) s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

From rest with constant acceleration, write $S(t) = K t^2$ where $K = a/2$.

H2 · MODEL

From rest with constant acceleration, write $S(t) = K t^2$ where $K = a/2$. Then $S_1 = K(p-1)^2$ and $S_2 = Kp^2$.

H3 · START

From rest with constant acceleration, write $S(t) = K t^2$ where $K = a/2$. Then $S_1 = K(p-1)^2$ and $S_2 = Kp^2$. If the same body covers S_1+S_2 in time T, then $K T^2 = K[(p-1)^2 + p^2]$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q22 EX-M6B-01

MEDIUM

M-ST06B · Two Braking Cars: Kinematics, Then Geometry

TWO CARS BRAKE TOWARDS EACH OTHER, JEE Main 2024, 9 Apr Evening - adapted

Two cars travel towards each other at 20 m/s each. When they are 300 m apart, both drivers apply brakes and each car retards at 2 m/s^2 . Find the distance between them when both come to rest.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Solve each stopping motion independently.

H2 · MODEL

Each stopping distance is $u^2/(2|a|) = 100 \text{ m}$.

H3 · START

Remaining gap = $300 - 100 - 100$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q23 EX-M6C-01

HARD

M-ST06C · Velocity Change + Average Velocity: A Short Route

DISPLACEMENT + VELOCITY INCREASE, JEE Main 2024, 30 Jan Morning - adapted

During the one-second interval from t to $t+1$, a particle has displacement 125 m and its velocity increases by 50 m/s. Find the distance travelled during the $(t+2)$ th second.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The given interval is exactly 1 s, so $\Delta v = 50$ immediately gives $a = 50 \text{ m/s}^2$.

H2 · MODEL

Average velocity in the given interval is 125 m/s.

H3 · START

Use $(u+v)/2 = 125$ and $v-u = 50$ to recover the endpoint velocities, then advance to the requested later second.

EXAMSIDE SOURCE

CHECK METHOD ->

Q24 EX-M7B-01

MEDIUM

M-ST07B · Free Fall from Rest

2026-04-05 DROPPED FROM 18 M

A ball is dropped from rest from 18 m. Find the height above ground where $|v|$ is numerically equal to $|g|$, using $g=10 \text{ m/s}^2$.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Use $v^2=2gs$ for the distance fallen, then subtract from 18 m.

H2 · MODEL

$v=10$, $u=0$. From $v^2=2gs$: $100=20s$, so $s=5 \text{ m}$ fallen.

H3 · START

$v=10$, $u=0$. From $v^2=2gs$: $100=20s$, so $s=5 \text{ m}$ fallen. Height above ground = $18-5 = 13 \text{ m}$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q25 EX-M7B-02

HARD

M-ST07B · Free Fall from Rest

2022-07-29 FIRST HALF VS SECOND HALF

A ball is released from height h . Compare the times t_1 and t_2 needed to cover the first and second halves of the distance.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

For free fall from rest, cumulative time is proportional to (distance fallen).

H2 · MODEL

Let total fall time be T . Time for the first half is $t_1=T/(2)$.

H3 · START

The second-half time is $t_2=T-t_1$; simplify the ratio $t_1:t_2$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q26 EX-M7B-03

HARD

M-ST07B · Free Fall from Rest

2021-08-27 SHOWER DROPS

Drops fall regularly from 9.8 m. When the first drop hits the floor, the third just starts. Locate the second drop from the floor.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Find first-drop fall time, split it into two equal release intervals, then use $s = \frac{1}{2}gt^2$ for the second drop.

H2 · MODEL

From $h=9.8$ m and $g=9.8$, first-drop time $T=2$ s. If the third just begins when the first hits, the release gap is $T/2$.

H3 · START

From $h=9.8$ m and $g=9.8$, first-drop time $T=2$ s. If the third just begins when the first hits, the release gap is $T/2$. The second drop has fallen for $T/2$, so distance fallen $= \frac{1}{2} g(T/2)^2 = 2.45$ m.

EXAMSIDE SOURCE

CHECK METHOD ->

Q27 EX-M7C-01

HARD

M-ST07C · Last-Distance and Between-Point Fall Problems

2024-01-27 BODY CROSSES A AND B

A body falling under gravity covers points A and B separated by 80 m in 2 s. Find the distance of A from the starting point, using $g=10$.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Let A be reached at T and B at $T+2$. Use $S(T+2)-S(T)=80$, then compute $S(T)$.

H2 · MODEL

$S(t)=5t^2$. Use $5[(T+2)^2-T^2]=80 \rightarrow 5(4T+4)=80 \rightarrow T=3$ s.

H3 · START

$S(t)=5t^2$. Use $5[(T+2)^2-T^2]=80 \rightarrow 5(4T+4)=80 \rightarrow T=3$ s. Therefore distance of A from release $= S(3)=5 \cdot 9=45$ m.

EXAMSIDE SOURCE

CHECK METHOD ->

Q28 EX-M8A-01

EASY

M-ST08A · Highest Point: $v = 0$, $a = -g$

VERTICAL VELOCITY AT 3 s AND 5 s JEE Main 2023, 12 Apr Morning - adapted

A ball is thrown vertically upward at 150 m/s. With $g=10 \text{ m/s}^2$, the ratio of its velocities after 3 s and 5 s is written as $(x+1)/x$. Find x .

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Upward positive $\rightarrow a=-g$. Both times are before the top.

H2 · MODEL

$v(3)=120$ and $v(5)=100$, so the ratio is $6/5$.

H3 · START

$v_3=150-10 \cdot 3=120 \text{ m/s}$ and $v_5=150-10 \cdot 5=100 \text{ m/s}$. Ratio = $120/100=6/5=(x+1)/x$, giving $x=5$.

EXAMSIDE SOURCE

CHECK METHOD $\rightarrow$

Q29 EX-M8B-01

MEDIUM

M-ST08B · Time to Top, Maximum Height, Flight Time

2022-07-29 JUGGLER

A juggler throws n balls per second and releases each new ball when the previous one reaches its highest point. Find the maximum height.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Set $t_{up}=1/n$, then recover u and H .

H2 · MODEL

Time to top = $1/n$. Thus $u/g=1/n \rightarrow u=g/n$.

H3 · START

Time to top = $1/n$. Thus $u/g=1/n \rightarrow u=g/n$. Then $H=u^2/(2g)=g/(2n^2)$.

EXAMSIDE SOURCE

CHECK METHOD $\rightarrow$

Q30 EX-M8B-02

HARD

M-ST08B · Time to Top, Maximum Height, Flight Time

2022-07-28 BALL FROM TOWER

A ball is thrown upward from a tower at 19.6 m/s and hits the ground after 6 s. Find the greatest height above ground reached by the ball.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

First use the 6 s landing condition to get tower height; then add the rise $u^2/(2g)$.

H2 · MODEL

Take upward positive and launch point $y=0$. At ground after 6 s, $y=-h$.

H3 · START

Take upward positive and launch point $y=0$. At ground after 6 s, $y=-h$. So $-h=19.6(6)-4.9(36)=117.6-176.4=-58.8$, hence tower height $h=58.8$ m.

EXAMSIDE SOURCE

CHECK METHOD ->

Q31 EX-M8C-01

HARD

M-ST08C · Same Height Twice

HEIGHT $h/3$ ON THE WAY UP AND DOWN JEE Main 2022, 29 Jul Morning - adapted

A ball is thrown vertically upward and reaches maximum height h . Find the ratio of the times at which it is at height $h/3$ while going up and coming down.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Use $H=u^2/(2g)$, then solve $y=h/3$ in $y=ut-\frac{1}{2}gt^2$. The two positive roots are the two passage times.

H2 · MODEL

At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-\frac{1}{2}gt^2$.

H3 · START

At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-\frac{1}{2}gt^2$. Multiply by 6g: $u^2=6gut-3g^2t^2$.

EXAMSIDE SOURCE

CHECK METHOD ->

Q32 EX-M9A-01

HARD

M-ST09A · Relative Acceleration: Common Gravity Cancels

JEE MAIN 2022, 26 JUN EVENING

A ball is projected upward at 50 m/s at $t=0$. Another is projected upward at 50 m/s at $t=2$ s. Find the meeting time, $g=10$.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

At $t=2$, Ball 1 is 80 m above launch and moving at 30 m/s. Ball 2 starts at 50 m/s. After that, $a_{rel}=0$ and closing speed is 20 m/s.

H2 · MODEL

At $t=2$ s, Ball 1: $y=50(2)-5(4)=80$ m and $v=50-20=30$ m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and $a_{rel}=0$.

H3 · START

At $t=2$ s, Ball 1: $y=50(2)-5(4)=80$ m and $v=50-20=30$ m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and $a_{rel}=0$. Time after second launch $=80/20=4$ s.

EXAMSIDE SOURCE

CHECK METHOD ->

Q33 EX-M9B-01

MEDIUM

M-ST09B · Dropped from a Moving Platform

STONE RELEASED FROM RISING BALLOON JEE Main 2026, 8 Apr Evening - adapted

A gas balloon rises at 10 m/s. At height 75 m a stone is dropped, while the balloon continues upward at the same speed. Find the balloon's height when the stone hits the ground, taking $g=10$.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Stone starts with $u=+10$ m/s in the ground frame. Solve $0=75+10t-5t^2$; then advance the balloon by $10t$.

H2 · MODEL

Use upward positive. Stone: $y=75+10t-5t^2$.

H3 · START

Use upward positive. Stone: $y=75+10t-5t^2$. Setting $y=0$ gives $t=5$ s.

EXAMSIDE SOURCE

CHECK METHOD ->

Q34 EX-M9C-01

MEDIUM

M-ST09C · Upward vs Downward Launch: Same Impact Speed

TWO BALLS FROM A BUILDING AIEEE 2002 - adapted

From a building, Ball A is thrown upward and Ball B downward with the same speed magnitude. Compare their velocities on reaching the ground, neglecting air resistance.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Their velocity directions at impact are both downward. For speed magnitude, use $v^2 = u^2 + 2gH$ for each.

H2 · MODEL

Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2 = u^2 + 2gH$.

H3 · START

Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2 = u^2 + 2gH$. Therefore the impact speed magnitudes are equal.

EXAMSIDE SOURCE

CHECK METHOD ->

Q35 EX-M9D-02

MEDIUM

M-ST09D · Time Gap: Upward vs Downward Launch from a Tower

JEE MAIN 2022, 24 JUN MORNING

From a tower, upward launch reaches ground in 6 s and equal-speed downward launch in 1.5 s. Find the drop-from-rest time.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

This is the paired-time tower case: upward time 6 s and downward time 1.5 s.

H2 · MODEL

Use $t_0 = (t_1 t_2) = (6 \times 1.5)$.

H3 · START

(9) = 3 s.

EXAMSIDE SOURCE

CHECK METHOD ->

Q36 EX-M10A-01

HARD

M-ST10A · Position-Time Graph: Slope Is Velocity

JEE MAIN 2025, 4 APR EVENING

From the supplied displacement-time graph, explain how to obtain average velocity and instantaneous velocity using secant and tangent slopes.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

On an x-t graph, height is position and slope is velocity.

H2 · MODEL

Average velocity over an interval comes from the secant slope $\Delta x / \Delta t$; instantaneous velocity comes from the tangent slope.

H3 · START

Test each statement using the appropriate secant or tangent slope, then select the supported statements.

EXAMSIDE SOURCE

CHECK METHOD ->

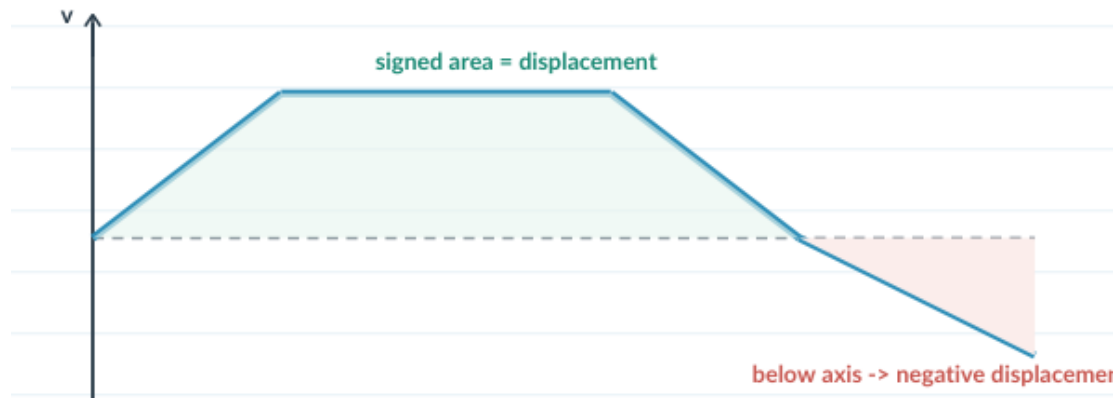
Q37 EX-M10B-02

MEDIUM

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

JEE MAIN 2025, 28 JAN EVENING

Find distance from $t=0$ to 4 s from a v-t graph.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Distance from a v-t graph is the area under the curve when velocity stays positive.

H2 · MODEL

From 0-2 s use the triangle; from 2-4 s use the rectangle.

H3 · START

Area = 10 + 20 m.

EXAMSIDE SOURCE

CHECK METHOD ->

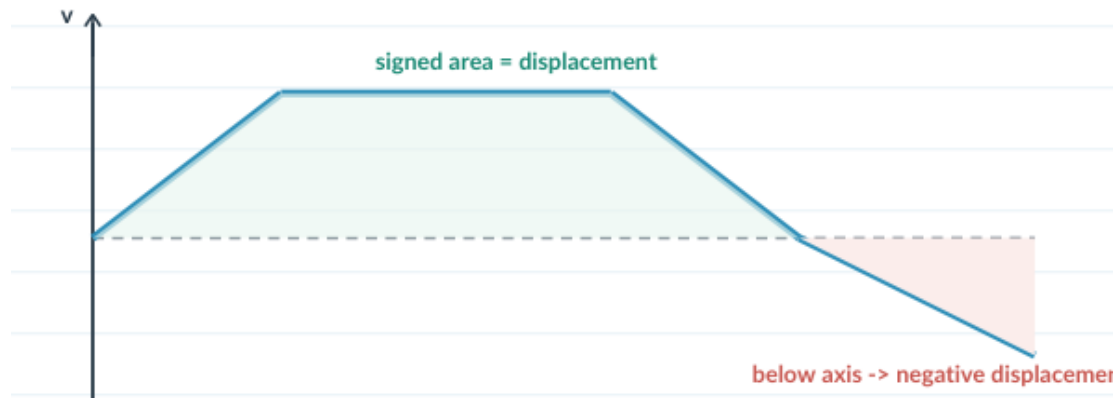
Q38 EX-M10B-03

MEDIUM

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

JEE MAIN 2025, 23 JAN MORNING

Find airplane distance in the first 30.5 s from its v-t graph.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Treat each simple part of the v-t graph as a geometric area.

H2 · MODEL

Add the areas over the first 30.5 s; use the graph units consistently.

H3 · START

The source route sums the regions to 12 km.

EXAMSIDE SOURCE

CHECK METHOD ->

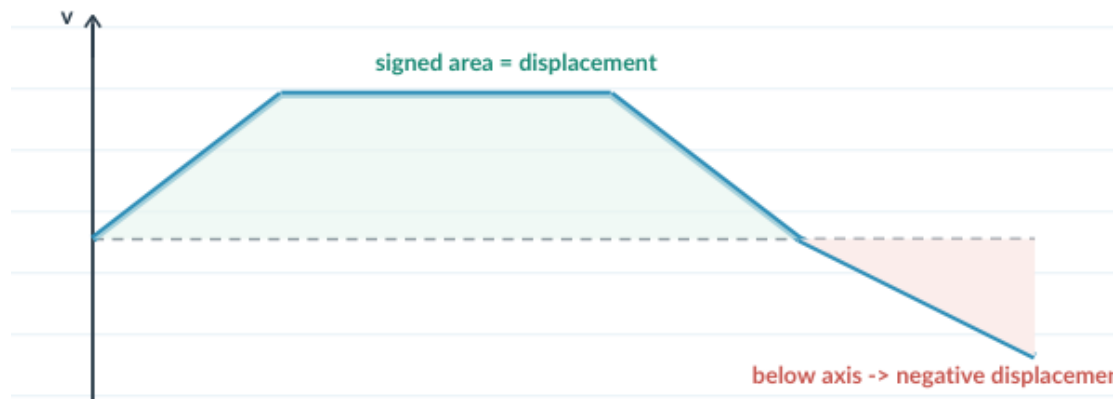
Q39 EX-M10B-05

MEDIUM

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

JEE MAIN 2023, 8 APR EVENING

Judge: area under v-t gives distance; area under a-t gives change in velocity.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Remember: v-t signed area is displacement, while a-t area is change in velocity.

H2 · MODEL

Check Statement I against displacement versus distance; check Statement II against Δv .

H3 · START

The source-supported choice is 'I false, II true'.

EXAMSIDE SOURCE

CHECK METHOD ->

Q40 EX-M10B-01

HARD

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

JEE MAIN 2026, 5 APR EVENING

From a v-t graph over 40 s, find total distance and average velocity.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

On a v-t graph, signed area is displacement; absolute area is distance.

H2 · MODEL

Break the 40 s graph into simple triangular/rectangular regions and keep track of signs.

H3 · START

Add absolute areas for distance; add signed areas for displacement; then average velocity = displacement/40 s.

EXAMSIDE SOURCE

CHECK METHOD ->

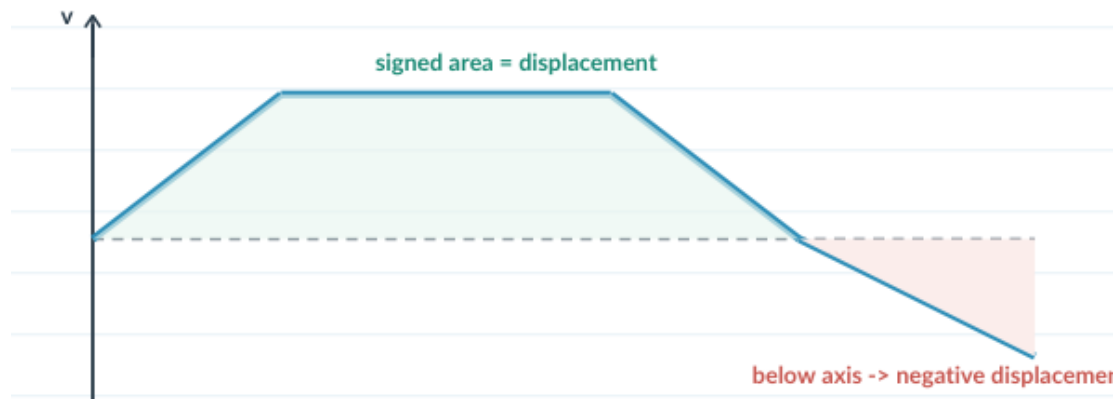
Q41 EX-M10B-04

HARD

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

JEE MAIN 2023, 24 JAN EVENING

Find displacement : distance from a v-t graph over 0-10 s.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Signed v-t area gives displacement; absolute area gives distance.

H2 · MODEL

Compute the positive and negative regions separately before combining them.

H3 · START

The source route gives signed area 16 m and absolute area 48 m; compare 16:48.

EXAMSIDE SOURCE

CHECK METHOD ->

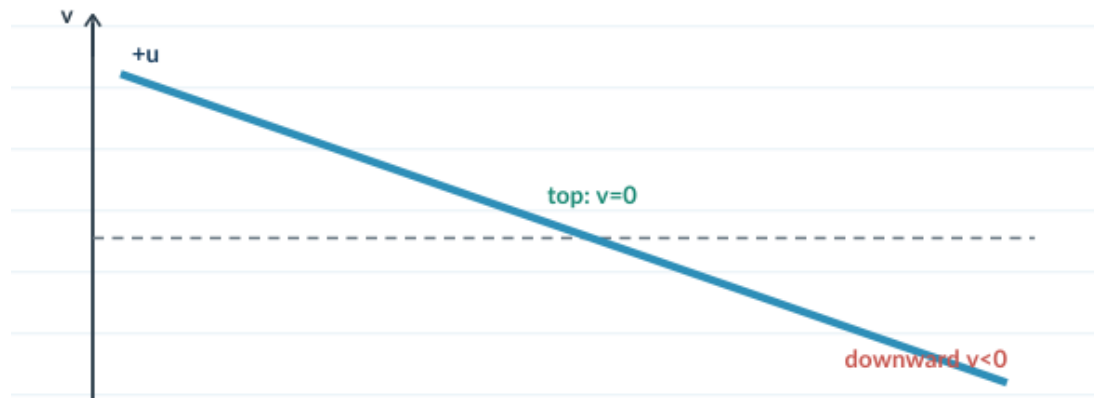
Q42 EX-M10C-01

MEDIUM

M-ST10C · Vertical Projection on a v-t Graph

JEE MAIN 2017 OFFLINE

Sketch or describe the v-t graph for a body thrown vertically upward.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

For a vertical throw, acceleration is constant $-g$, so the v-t graph must be a straight line.

H2 · MODEL

Use $v = u - gt$: the line crosses $v=0$ at the top and then continues to negative velocity.

H3 · START

Choose the graph with constant negative slope through zero and into negative v.

EXAMSIDE SOURCE

CHECK METHOD ->

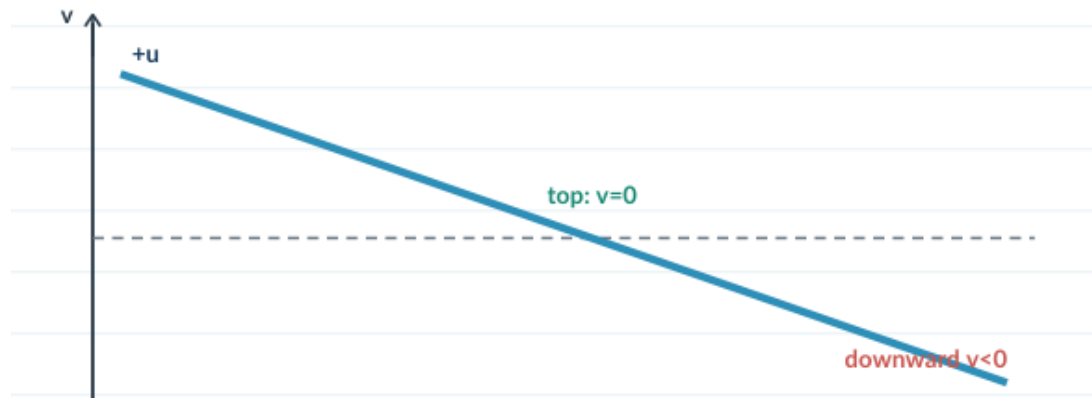
Q43 EX-M10C-02

HARD

M-ST10C · Vertical Projection on a v-t Graph

JEE MAIN 2022, 27 JUL MORNING

Sketch or describe the v-t curve for a bullet moving downward for 10 s, hitting the ground inelastically, then remaining at rest.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Use one sign convention consistently; the source method takes downward as positive.

H2 · MODEL

Before impact, v increases linearly from 100 m/s with slope $+g$.

H3 · START

At 10 s the speed is 200 m/s; the inelastic impact makes v jump to 0 and then remain 0.

EXAMSIDE SOURCE

CHECK METHOD ->

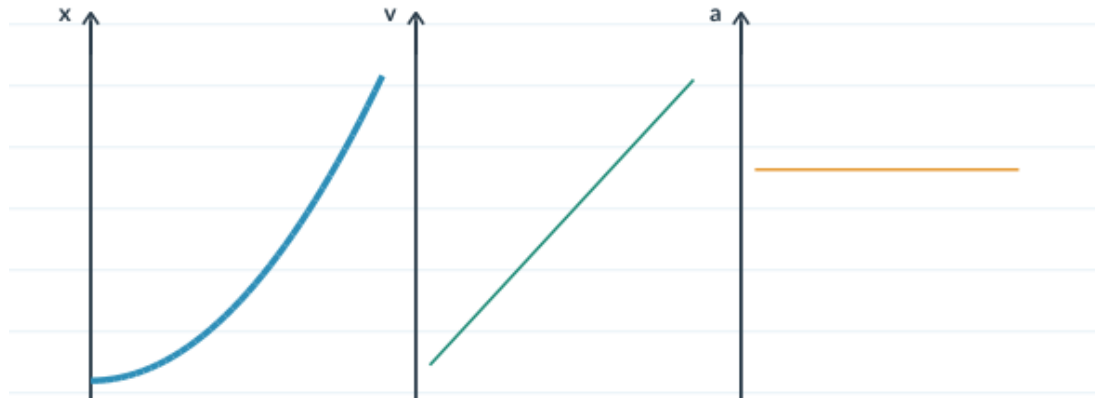
Q44 EX-M10D-01

MEDIUM

M-ST10D · Constant Acceleration: x-t Parabola, v-t Line, a-t Constant

JEE MAIN 2021, 18 MAR MORNING

Sketch or describe the x-t, v-t and a-t graph shapes for motion with constant acceleration.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Constant acceleration means a-t is horizontal.

H2 · MODEL

Then v-t is linear and x-t is quadratic/parabolic.

H3 · START

Choose the set showing x-t parabolic, v-t straight and a-t horizontal.

EXAMSIDE SOURCE

CHECK METHOD ->

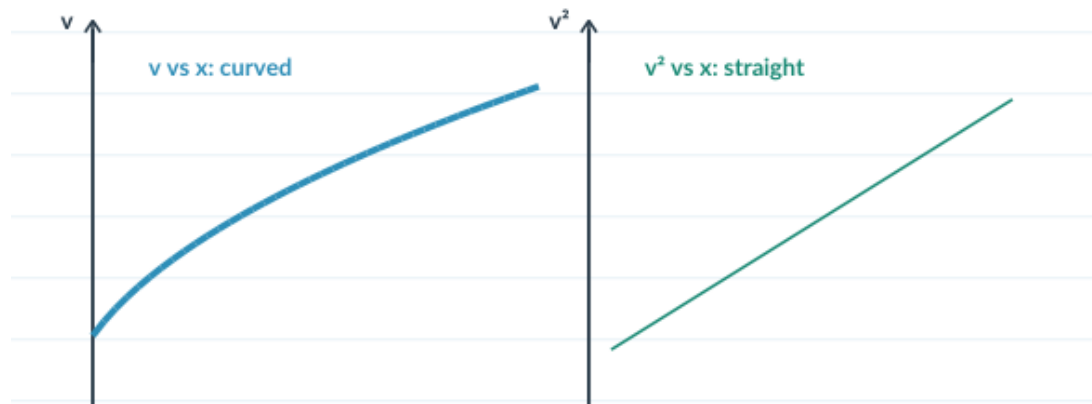
Q45 EX-M10E-03

MEDIUM

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

JEE MAIN 2021, 31 AUG EVENING

A v^2 - x graph is a straight line. Find constant acceleration from its slope.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

For constant acceleration, $v^2 = u^2 + 2ax$ is linear in x .

H2 · MODEL

Therefore the slope of a v^2 - x graph equals $2a$.

H3 · START

Given slope 2, solve $2a=2$.

EXAMSIDE SOURCE

CHECK METHOD ->

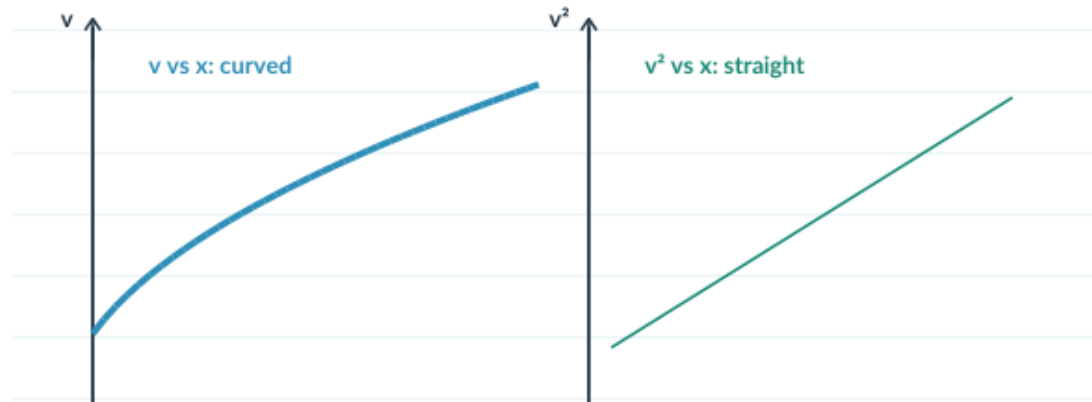
Q46 EX-M10E-04

HARD

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

JEE MAIN 2017, 8 APR MORNING

For positive velocity with constant negative acceleration, sketch or describe the v-x graph.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Use $v^2 = u^2 + 2ax$ with $a < 0$.

H2 · MODEL

As x increases, v^2 decreases linearly, but v itself follows a square-root curve.

H3 · START

Choose the positive- v graph that decreases nonlinearly with x .

EXAMSIDE SOURCE

CHECK METHOD ->

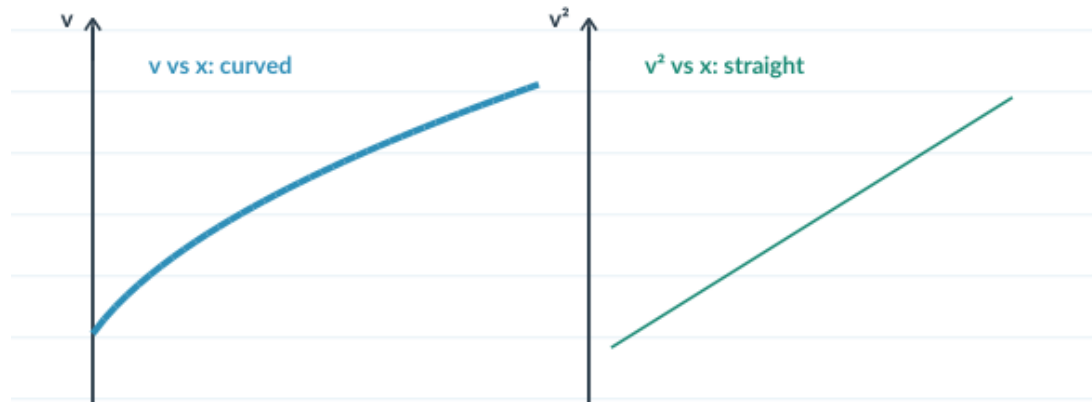
Q47 EX-M10E-01

CHALLENGE

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

JEE MAIN 2026, 24 JAN EVENING

For the given straight-line v - x graph, sketch or describe the corresponding a - x graph.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

A straight v - x relation is not itself constant acceleration.

H2 · MODEL

Use the bridge $a = v \, dv/dx$. For $v=c-mx$, $dv/dx=-m$.

H3 · START

Then $a=-m(c-mx)$: a straight line with positive slope and negative intercept.

EXAMSIDE SOURCE

CHECK METHOD ->

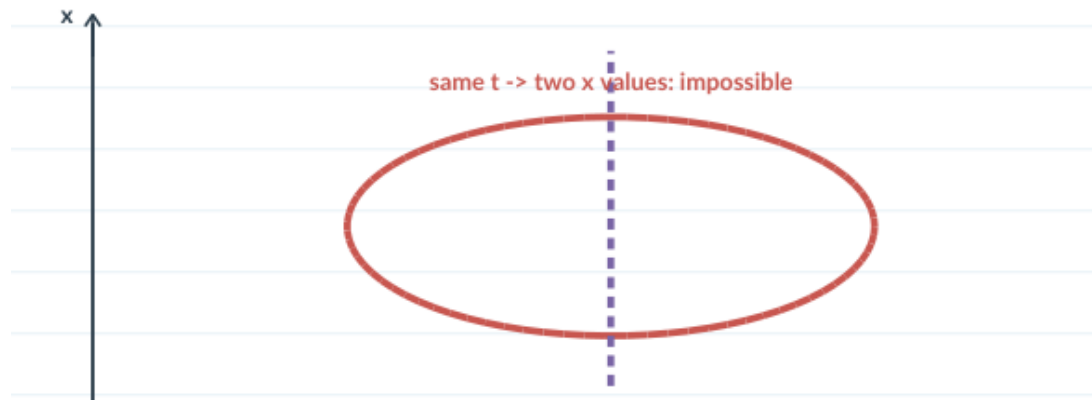
Q48 EX-M10F-01

CHALLENGE

M-ST10F · Impossible and Inconsistent Motion Graphs

JEE MAIN 2025, 3 APR MORNING

State the single-valued-time test used to decide whether a plotted curve can represent one-dimensional motion. The original option plots are not preserved in the audited source PDF.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

For one particle, one instant of time cannot correspond to two positions or two velocities.

H2 · MODEL

Check each plotted curve for whether time gives a single-valued state.

H3 · START

Reject the curve that assigns two values at the same t ; compare the remaining candidates.

EXAMSIDE SOURCE

CHECK METHOD ->

Q49 EX-M11D-01

HARD

M-ST11D · Ratio of nth-Second Distance to Total Distance

JEE MAIN 2024, 5 APR MORNING

For motion from rest, the ratio S_{n-1}/S_n of two consecutive one-second distances is written as $1-2/x$ for $n=10$. Find x .

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

For motion from rest with constant acceleration, nth-second distances are proportional to $2n-1$.

H2 · MODEL

So $s_9:s_{10} = 17:19$.

H3 · START

Set $17/19 = 1 - 2/x$ and solve for x .

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER • SET 1 / 2

MOTION IN A STRAIGHT LINE

Mixed transfer I • identify the concept before checking

Q50 EX-M2B-07

HARD

CONCEPT • IDENTIFY FIRST

JEE MAIN 2023 - 30 JAN MORNING

A horse rider covers half the distance at 5 m/s. The remaining half is covered at 10 m/s and 15 m/s for equal times. The mean speed is $x/7$ m/s. Find x .

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 • NOTICE

For the second half, equal times at 10 and 15 m/s give an equal-time average of 12.5 m/s.

H2 • MODEL

Let total distance be D . First-half time = $D/10$; second-half time = $(D/2)/12.5 = D/25$.

H3 • START

Average = $D/(D/10 + D/25) = 50/7$ m/s, so compare with $x/7$.

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER · SET 1 / 2

MOTION IN A STRAIGHT LINE

Mixed transfer I · identify the concept before checking

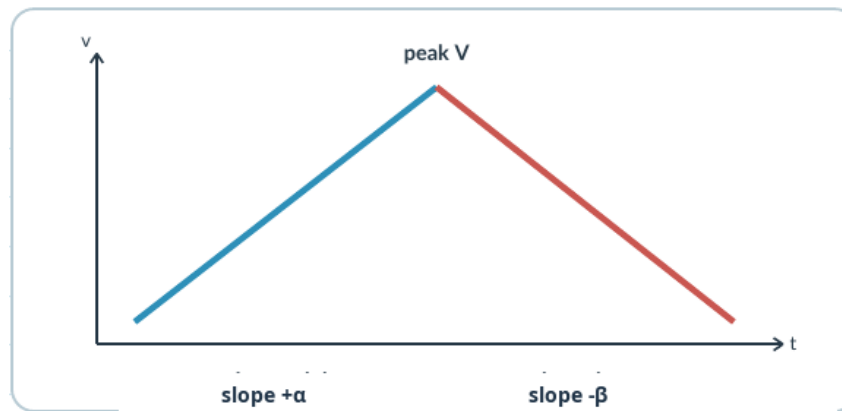
Q51 EX-M4C-02

CHALLENGE

CONCEPT · IDENTIFY FIRST

ACCELERATE, THEN BRAKE TO REST JEE Main 2021, 17 Mar Morning - adapted

A car accelerates from rest at constant rate α for some time, then decelerates at constant rate β to come to rest. If total time is T , express the total distance in terms of α , β and T .



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Both stages share the same peak velocity V ; the v - t graph is one triangle.

H2 · MODEL

$t_1 = V/\alpha$ and $t_2 = V/\beta$, so $T = V(1/\alpha + 1/\beta)$.

H3 · START

Solve for V , then total distance is the triangle area: $S = \frac{1}{2} V T$.

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER · SET 1 / 2

MOTION IN A STRAIGHT LINE

Mixed transfer I · identify the concept before checking

Q52 EX-M4E-04

HARD

CONCEPT · IDENTIFY FIRST

JEE MAIN 2022 - 25 JUN EVENING

Two buses start together with positions $XP = \alpha t + \beta t^2$ and $XQ = f t - t^2$. Find when their velocities are equal.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Read each position polynomial using $x = x_0 + ut + (\frac{1}{2})at^2$.

H2 · MODEL

For P: $u = \alpha$, $a = 2\beta$. For Q: $u = f$, $a = -2$.

H3 · START

Set $\alpha + 2\beta t = f - 2t$ and solve for t .

EXAMSIDE SOURCE

CHECK METHOD ->

Q53 EX-M9A-02

CHALLENGE

CONCEPT • IDENTIFY FIRST

JEE MAIN 2021, 26 AUG MORNING

Two equal spherical balls are thrown upward 3 s apart with the same initial velocity 35 m/s. Each radius is 5 cm. Find the collision height, $g=10$.

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 • NOTICE

At the second launch, compute the first ball's state.

H2 • MODEL

At the second launch (3 s), Ball 1 is at 60 m and moving upward at 5 m/s. Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant.

H3 • START

Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant. Allowing for the 10 cm centre separation at contact gives a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m.

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER · SET 1 / 2

MOTION IN A STRAIGHT LINE

Mixed transfer I · identify the concept before checking

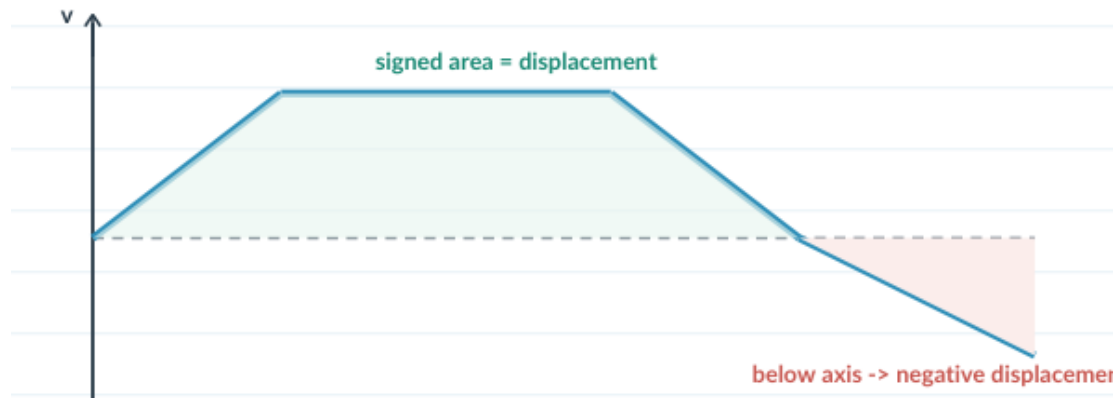
Q54 EX-M10B-06

HARD

CONCEPT · IDENTIFY FIRST

JEE MAIN 2023, 11 APR MORNING

From a v-t graph, find distance/displacement ratio in 25 s.



WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Use absolute v-t area for distance and signed v-t area for displacement.

H2 · MODEL

Evaluate the regions over the 25 s interval before taking the ratio.

H3 · START

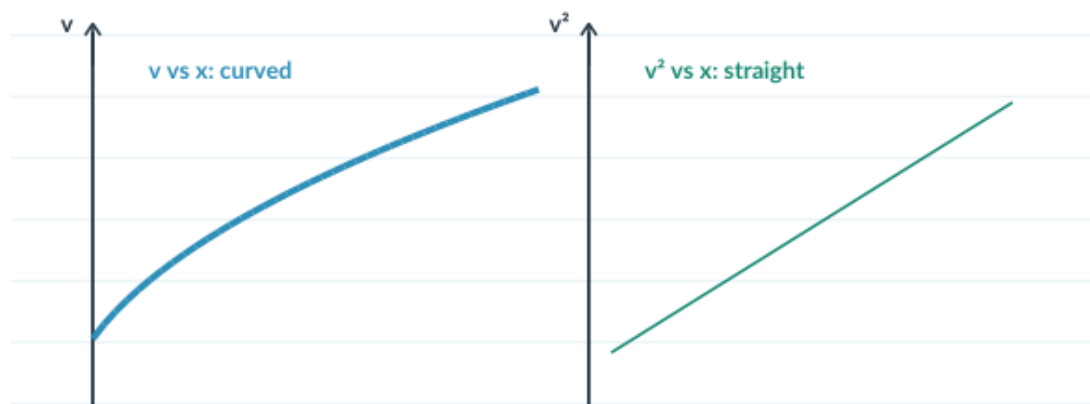
Distance = 250 m and displacement = 150 m, so form 250/150.

EXAMSIDE SOURCE

CHECK METHOD ->

Q55 EX-M10E-02**CHALLENGE****CONCEPT • IDENTIFY FIRST**

JEE MAIN 2021, 18 MAR EVENING

For a linear v - x graph, sketch or describe the corresponding a - x graph.

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.**H1 • NOTICE**Convert $v(x)$ to $a(x)$ using $a = v \, dv/dx$.**H2 • MODEL**For a linear v - x relation, dv/dx is constant, so a is proportional to v and is linear in x .**H3 • START**The source route gives a straight a - x line with positive slope and negative intercept.

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER • SET 2 / 2

Mixed transfer II • identify the concept before checking

MOTION IN A STRAIGHT LINE

Q56 EX-M4B-03

HARD

CONCEPT • IDENTIFY FIRST

TRANSFER QUESTION JEE Main 2024, 29 Jan Morning

A body starts from rest with constant acceleration. It covers S_1 in the first $(p-1)$ s and S_2 in the first p s. In what time is displacement $S_1 + S_2$ covered?

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 • NOTICE

From rest with constant a , cumulative displacement has the form Kt^2 .

H2 • MODEL

$S_1 = K(p-1)^2$ and $S_2 = Kp^2$. Their sum must equal KT^2 .

H3 • START

SET UP THE EQUATION $T^2 = (p-1)^2 + p^2 = 2p^2 - 2p + 1$. Take the positive square root for time.

EXAMSIDE SOURCE

CHECK METHOD ->

Q57 EX-M4D-04

HARD

CONCEPT • IDENTIFY FIRST

SPEED AT THE TRAIN MIDPOINT JEE Main 2021, 25 Feb Morning - adapted

A train accelerates uniformly. Its engine passes a signal with speed u , and the last carriage passes the same signal with speed v . What is the speed w when the midpoint of the train passes the signal?

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 • NOTICE

The midpoint corresponds to half the displacement from engine to last carriage.

H2 • MODEL

For full length L : $v^2 = u^2 + 2aL$.

H3 • START

For half length $L/2$: $w^2 = u^2 + aL$. Eliminate aL using the full-length equation.

EXAMSIDE SOURCE

CHECK METHOD ->

Q58 EX-M7C-02

CHALLENGE

CONCEPT · IDENTIFY FIRST

2021-07-25 SPACING BETWEEN DROPLETS

Water droplets fall regularly. The spacing between a droplet observed at its 4th second of fall and the next droplet is 34.3 m. Determine the drop rate, using $g=9.8$.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

Write spacing as difference of two free-fall distances with elapsed times separated by the release interval.

H2 · MODEL

Let release interval be τ . The older drop has fallen 4 s; the next one has fallen $4-\tau$ s.

H3 · START

Let release interval be τ . The older drop has fallen 4 s; the next one has fallen $4-\tau$ s. Spacing = $4.9[16-(4-\tau)^2]=34.3$.

EXAMSIDE SOURCE

CHECK METHOD ->

MIXED TRANSFER · SET 2 / 2

MOTION IN A STRAIGHT LINE

Mixed transfer II · identify the concept before checking

Q59 EX-M9D-01

HARD

CONCEPT · IDENTIFY FIRST

JEE MAIN 2024, 6 APR EVENING

A body thrown upward from a tower reaches ground in t_1 . Thrown downward with the same speed it reaches in t_2 . Find the time if simply dropped.

WORK HERE · TRY BEFORE LOOKING DOWN

STOP · LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 · NOTICE

The upward and downward launches use the same tower height and the same speed magnitude.

H2 · MODEL

Use the paired tower equations, or the source-derived relation connecting the two launch times.

H3 · START

For the drop-from-rest time, use $t_0 = (t_1 \ t_2)$.

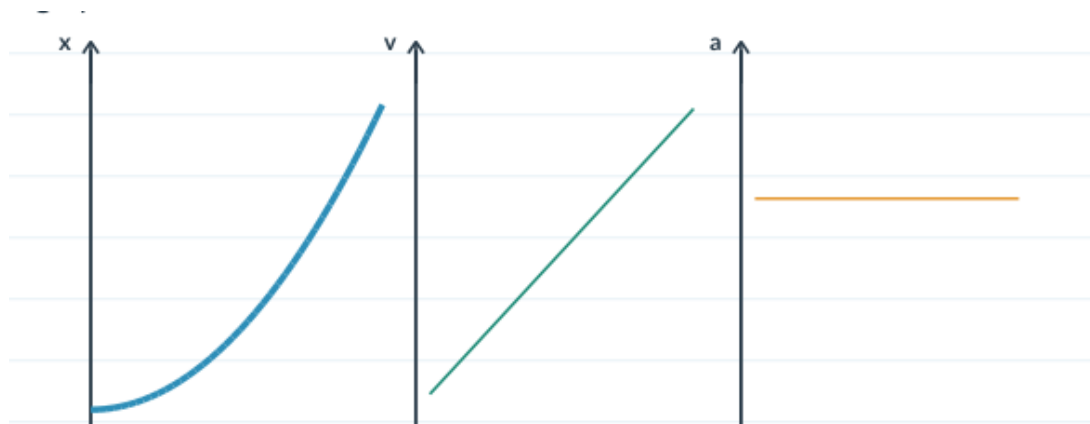
EXAMSIDE SOURCE

CHECK METHOD ->

Q60 EX-M10D-02**HARD**

CONCEPT • IDENTIFY FIRST

JEE MAIN 2019, 8 APR EVENING

From rest with uniform positive acceleration, identify all correct qualitative graphs.

WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.**H1 • NOTICE**

The particle starts from rest with constant positive acceleration.

H2 • MODELUse $v=at$ and $x=(\frac{1}{2})at^2$ to infer the graph shapes.**H3 • START**Check which candidates show linear v , quadratic x and constant positive a .

EXAMSIDE SOURCE

CHECK METHOD ->

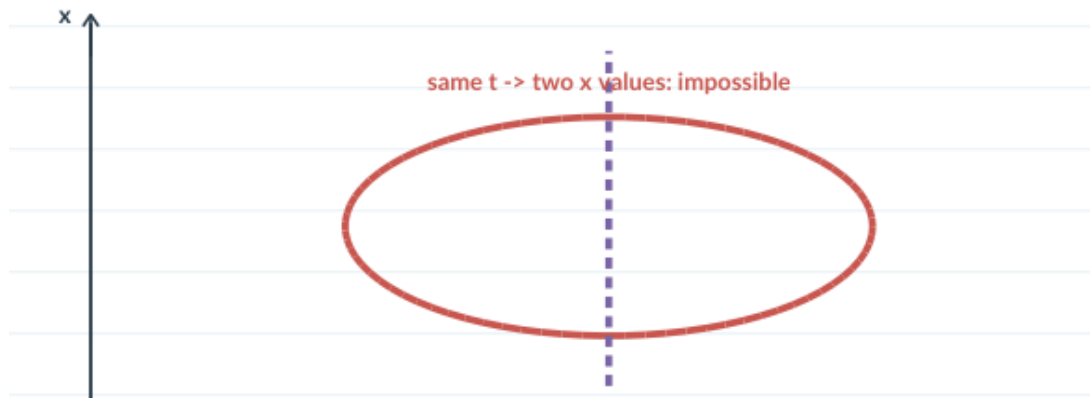
Q61 EX-M10F-02

CHALLENGE

CONCEPT • IDENTIFY FIRST

JEE MAIN 2018 OFFLINE

State the consistency test for multiple graph representations of the same motion. The audited source records graph 3 as the inconsistent source option.



WORK HERE • TRY BEFORE LOOKING DOWN

STOP • LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 • NOTICE

Translate every candidate graph into the same motion story before comparing them.

H2 • MODEL

Compare initial position and especially initial slope, because $x-t$ slope is velocity.

H3 • START

The inconsistent graph is the one whose initial slope implies zero initial velocity while the others do not.

EXAMSIDE SOURCE

CHECK METHOD ->

CHECK YOUR METHOD

Concept-assimilating solutions · use only after attempting

QUESTION RECAP -> required representation -> WHY THIS WORKS -> METHOD -> ANSWER / CHECK -> CONCEPT TO KEEP

The difficulty badge stays learner-friendly.

The canonical concept is always revealed here, including questions that were mixed-transfer in the practice book.

Q1 EX-M2B-02

EASY

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A person travels distance x at velocity v_1 and then another equal distance x at velocity v_2 in the same direction. Relate the average velocity v to v_1 and v_2 .

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each leg have distance x . The travel times are x/v_1 and x/v_2 , so total distance = $2x$ and total time = $x/v_1 + x/v_2$. Therefore $v_{\text{avg}} = 2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

ANSWER / CHECK

$$v = 2v_1v_2/(v_1+v_2)$$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

Q2 EX-M2B-04

EASY

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A vehicle travels 4 km at 3 km/h and another 4 km at 5 km/h. Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Do not average 3 and 5 directly because the legs have equal distance, not equal time. The times are $4/3$ h and $4/5$ h. Thus total time = $32/15$ h and $v_{\text{avg}} = 8/(32/15) = 3.75$ km/h.

ANSWER / CHECK

3.75 km/h

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

Q3 EX-M2B-05

EASY

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A car travels distance x at speed v_1 and then the same distance x at speed v_2 . Find average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each leg be x . Total distance = $2x$, while total time = $x/v_1 + x/v_2$. Hence $v_{avg} = 2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

ANSWER / CHECK

$$v_{avg} = 2v_1v_2/(v_1+v_2)$$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

[<- RETURN TO QUESTION](#)

Q4 EX-M2B-03

MEDIUM

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

An object covers three equal consecutive distances at speeds v_1 , v_2 and v_3 . Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each equal segment be L . The three times are L/v_1 , L/v_2 and L/v_3 , so total time = $L(1/v_1 + 1/v_2 + 1/v_3)$ and total distance = $3L$. Dividing gives $v_{avg} = 3/(1/v_1 + 1/v_2 + 1/v_3)$.

ANSWER / CHECK

$$v_{avg} = 3/(1/v_1 + 1/v_2 + 1/v_3)$$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

[<- RETURN TO QUESTION](#)

Q5 EX-M2B-06**MEDIUM**

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A person moves distance x at 5 m/s and the next $3x/2$ at speed v_2 . The overall average velocity is $50/7$ m/s. Find v_2 .

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Write time for each leg as distance/speed. Total displacement = $x + 3x/2 = 5x/2$, while total time = $x/5 + 3x/(2v_2)$. Set average velocity = $(5x/2)/(x/5 + 3x/(2v_2)) = 50/7$; cancel x and solve to get $v_2 = 10$ m/s.

ANSWER / CHECK $v_2 = 10$ m/s**CONCEPT TO KEEP**

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q6 EX-M2B-08**MEDIUM**

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A car covers three equal one-third portions at 11, 22 and 33 m/s. Find average velocity.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Use one common distance L for each third. Total distance = $3L$ and total time = $L/11 + L/22 + L/33$. Therefore $v_{av} = 3/(1/11 + 1/22 + 1/33) = 18$ m/s.

ANSWER / CHECK

18 m/s

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q7 EX-M2B-01**HARD**

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

Half the journey is at 6 m/s. The remaining half is covered in two equal time intervals at 9 m/s and 15 m/s. Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Treat the second half first: equal times at 9 m/s and 15 m/s give 12 m/s for that half. Let each half-distance be L . Then the first-half time is $L/6$ and the second-half time is $L/12$. Therefore $v_{\text{avg}} = 2L/(L/6 + L/12) = 8 \text{ m/s}$.

ANSWER / CHECK

8 m/s

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q8 EX-M3A-01**HARD**

M-ST03A · Acceleration as Change in Velocity

QUESTION RECAP

A tennis ball is dropped from 9.8 m and rebounds to 5.0 m. Contact with the floor lasts 0.2 s. Take $g=10 \text{ m/s}^2$. Find average acceleration during contact.

WHY THIS WORKS

Acceleration depends on change in velocity, so direction matters. A rebound reverses the velocity sign and makes the velocity change larger than the change in speed alone.

METHOD

Choose upward positive. Just before impact $v=-14 \text{ m/s}$; just after rebound $v=+10 \text{ m/s}$. $\Delta v=24 \text{ m/s}$ in $0.2 \text{ s} \rightarrow$ average acceleration= 120 m/s^2 upward.

ANSWER / CHECK

$\Delta v=24 \text{ m/s}$ in $0.2 \text{ s} \rightarrow$ average acceleration= 120 m/s^2 upward

CONCEPT TO KEEP

Keep: average acceleration = $\Delta v/\Delta t$, with velocity signs included.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q9 EX-M4A-01**EASY****M-ST04A · Constant Acceleration and $v = u + at$** **QUESTION RECAP**

A particle moves in +x with $u = 10 \text{ m/s}$ and constant $a = 2 \text{ m/s}^2$. How long to reach 60 m/s ?

WHY THIS WORKS

Here distance is not needed: the known quantities are initial velocity, final velocity and constant acceleration, so the direct velocity-time relation is the efficient model.

METHOD

$u = 10$, $v = 60$, $a = +2$. Use $v = u + at$: $60 = 10 + 2t \rightarrow t = 25 \text{ s}$. Check: positive acceleration increases velocity.

ANSWER / CHECK

25 s

CONCEPT TO KEEP

Keep: if u , v , a and t are the useful variables under constant acceleration, $v = u + at$ is the direct relation.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q10 EX-M4B-01**MEDIUM****M-ST04B · Displacement under Constant Acceleration****QUESTION RECAP**

A bus moving at 72 km/h is brought to rest uniformly in 4 s . Find the braking distance.

WHY THIS WORKS

Uniform braking is one constant-acceleration interval. First recover the acceleration from the velocity change, then use it consistently to find the distance.

METHOD

Convert $72 \text{ km/h} = 20 \text{ m/s}$. With $v = 0$ and $t = 4 \text{ s}$, $0 = 20 + 4a \rightarrow a = -5 \text{ m/s}^2$. Then $s = ut + \frac{1}{2}at^2 = 20(4) + \frac{1}{2}(-5)(16) = 80 - 40 = 40 \text{ m}$.

ANSWER / CHECK

40 m

CONCEPT TO KEEP

Keep: one constant-acceleration model must be used consistently through a uniform braking interval.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

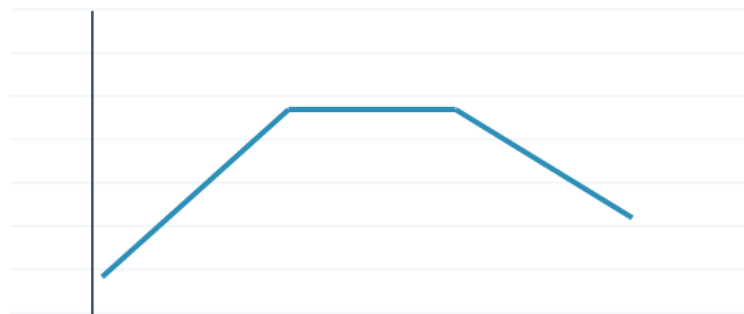
Q11 EX-M4B-04

MEDIUM

M-ST04B · Displacement under Constant Acceleration

QUESTION RECAP

A particle starts at the origin and moves along +x with the shown v-t graph. Find its position at $t = 5$ s.



WHY THIS WORKS

On a velocity-time graph, signed area gives displacement. Because the particle starts at the origin, its final position equals that accumulated signed area.

METHOD

The particle starts from $x = 0$, so its position after 5 s equals the signed area under the given v-t graph from 0 to 5 s. Break the graph into its simple pieces and add their areas. ExamSIDE gives the resulting position as 9 m.

ANSWER / CHECK

the resulting position as 9 m

CONCEPT TO KEEP

Keep: signed area under v-t = displacement; add the initial position to obtain final position.

Q12 EX-M4C-01

MEDIUM

M-ST04C · Average Velocity under Constant Acceleration

QUESTION RECAP

A train starts from rest, accelerates uniformly to 80 km/h during time t , then continues at 80 km/h for another $3t$. Find the average speed for the whole $4t$ journey.

WHY THIS WORKS

A multistage journey must be handled stage by stage. The accelerating stage has a linearly changing velocity, while the cruising stage has constant velocity.

METHOD

Stage 1 is uniform acceleration from 0 to 80, so $\text{avg}_1 = 40$ km/h. Hence $s_1 = 40t$. Stage 2 is constant 80 km/h for $3t$, so $s_2 = 240t$. Total distance = $280t$ and total time = $4t$. Therefore average speed = 70 km/h.

ANSWER / CHECK

average speed = 70 km/h

CONCEPT TO KEEP

Keep: for staged motion, compute each stage distance and then use total distance / total time.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

Q13 EX-M4D-01

MEDIUM

M-ST04D · Remove Time: Why Velocities Become Squared

QUESTION RECAP

An automobile travelling at 40 km/h can be stopped in 40 m under a fixed braking condition. Under the same condition, what stopping distance is needed at 80 km/h?

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2 .

METHOD

For fixed braking acceleration, stopping distance $\propto u^2$. The speed ratio is $80/40 = 2$, so the stopping-distance ratio is $2^2 = 4$. Therefore the required distance is $4 \times 40 = 160$ m.

ANSWER / CHECK

160 m

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

Q14 EX-M4D-02

HARD

M-ST04D · Remove Time: Why Velocities Become Squared

QUESTION RECAP

A train moving at 20 m/s normally needs 500 m to stop under a fixed retardation. If braking begins only 250 m before the station, the train crosses the station at speed (x) m/s. Find x .

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2 .

METHOD

Full stop: $0 = 20^2 + 2a(500)$, so $1000a = -400$ and $a = -0.4 \text{ m/s}^2$. If braking starts 250 m before the station: $v^2 = 20^2 + 2(-0.4)(250) = 400 - 200 = 200$. Since $v = x$, $x = 200$.

ANSWER / CHECK

Since $v = x$, $x = 200$

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q15 EX-M4D-03

HARD

M-ST04D · Remove Time: Why Velocities Become Squared

QUESTION RECAP

A bullet's speed becomes one-third of its entry speed after penetrating 4 cm into a wooden block. Assume constant resistance. If it finally stops after travelling $(4 + x)$ cm in the block, find x .

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2 .

METHOD

Let the entry speed be u and constant retardation magnitude be b . Over the first 4 cm, the loss in v^2 is $u^2 - (u/3)^2 = 8u^2/9$. Over the remaining distance x , the loss is $(u/3)^2 - 0 = u^2/9$. Because the loss in v^2 per unit distance is constant, distance is proportional to the v^2 loss. Therefore $x/4 = (1/9)/(8/9) = 1/8$, giving $x = 0.5 \text{ cm}$.

ANSWER / CHECK

$x/4 = (1/9)/(8/9) = 1/8$, giving $x = 0.5 \text{ cm}$

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q16 EX-M4E-01**MEDIUM****M-ST04E · Read a Motion Equation from Its Coefficients****QUESTION RECAP**

$x(t)=5t^2-4t+5$ m. Find the magnitude of velocity at $t=2$ s without calculus.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + \frac{1}{2}at^2$. This reveals u and a without using calculus.

METHOD

Compare with $x=x_0+ut+\frac{1}{2}at^2$: $x_0=5$, $u=-4$, $a=10$. Then $v=u+at=-4+20=16$ m/s.

ANSWER / CHECK

16 m/s

CONCEPT TO KEEP

Keep: match a quadratic $x(t)$ to $x_0 + ut + \frac{1}{2}at^2$ to read u and a directly.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q17 EX-M4E-02**MEDIUM****M-ST04E · Read a Motion Equation from Its Coefficients****QUESTION RECAP**

$s(t)=2.5t^2$. Find instantaneous speed at $t=5$ s using the constant-acceleration form.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + \frac{1}{2}at^2$. This reveals u and a without using calculus.

METHOD

Read $u=0$ and $\frac{1}{2}a=2.5 \rightarrow a=5$ m/s². Then $v=at=25$ m/s.

ANSWER / CHECK

25 m/s

CONCEPT TO KEEP

Keep: match a quadratic $x(t)$ to $x_0 + ut + \frac{1}{2}at^2$ to read u and a directly.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q18 EX-M4E-03**MEDIUM**

M-ST04E · Read a Motion Equation from Its Coefficients

QUESTION RECAP

Distance from the start is $x=4t^2$. Find velocity at $t=5$ s under the constant-acceleration interpretation.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + \frac{1}{2}at^2$. This reveals u and a without using calculus.

METHOD

Read $u=0$ and $\frac{1}{2}a=4 \rightarrow a=8 \text{ m/s}^2$. Then $v=8t=40 \text{ m/s}$ at 5 s.

ANSWER / CHECK

40 m/s

CONCEPT TO KEEP

Keep: match a quadratic $x(t)$ to $x_0 + ut + \frac{1}{2}at^2$ to read u and a directly.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q19 EX-M5A-02**MEDIUM**M-ST05A · Distance During the n th Second**QUESTION RECAP**

A small toy starts from rest under constant acceleration. It travels 10 m in the first t seconds. How far does it travel in the next t seconds?

WHY THIS WORKS

For constant acceleration, n th-second questions are interval questions built from cumulative displacement. Subtracting nearby cumulative expressions removes quantities that are not needed.

METHOD

Starting from rest with constant acceleration, $S(t)=\frac{1}{2}at^2$. Given $S(t)=10 \text{ m}$. Then $S(2t)=\frac{1}{2}a(2t)^2 = 4S(t)=40 \text{ m}$. The question asks only for the next t -second interval, so subtract: $40 - 10 = 30 \text{ m}$.

ANSWER / CHECK

30 m

CONCEPT TO KEEP

Keep: n th-second displacement comes from subtracting cumulative displacements at adjacent times.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q20 EX-M5A-01**HARD**

M-ST05A · Distance During the nth Second

QUESTION RECAP

A body travels 102.5 m during its n th second and 115.0 m during its $(n+2)$ th second. Find its constant acceleration.

WHY THIS WORKS

For constant acceleration, n th-second questions are interval questions built from cumulative displacement. Subtracting nearby cumulative expressions removes quantities that are not needed.

METHOD

For constant acceleration, $s_n = u + (a/2)(2n-1)$. Two seconds later, $s_{n+2} = u + (a/2)(2n+3)$. Subtracting cancels u and n : $s_{n+2} - s_n = 2a$. Therefore $115.0 - 102.5 = 12.5 = 2a$, so $a = 6.25 \text{ m/s}^2$.

ANSWER / CHECK 6.25 m/s^2 **CONCEPT TO KEEP**

Keep: n th-second displacement comes from subtracting cumulative displacements at adjacent times.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q21 EX-M5B-01**HARD**

M-ST05B · Why Successive Distances Follow 1 : 3 : 5 : 7

QUESTION RECAP

A body starts from rest with constant acceleration. It covers S_1 in the first $(p-1)$ s and S_2 in the first p s. In what time is displacement $S_1 + S_2$ covered?

WHY THIS WORKS

Because the body starts from rest with constant acceleration, cumulative displacement has the common form Kt^2 . The common factor K cancels, so the problem is really about time-squared scaling.

METHOD

From rest with constant acceleration, write $S(t) = Kt^2$ where $K = a/2$. Then $S_1 = K(p-1)^2$ and $S_2 = Kp^2$. If $S_1 + S_2$ is covered in time T , then $KT^2 = K[(p-1)^2 + p^2]$. Cancel K to obtain $T^2 = 2p^2 - 2p + 1$, so $T = (2p^2 - 2p + 1) \text{ s}$.

ANSWER / CHECK $T = (2p^2 - 2p + 1) \text{ s}$ **CONCEPT TO KEEP**

Keep: when displacement from rest scales as t^2 , combine the squared times before taking the square root.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q22 EX-M6B-01**MEDIUM**

M-ST06B · Two Braking Cars: Kinematics, Then Geometry

QUESTION RECAP

Two cars travel towards each other at 20 m/s each. When they are 300 m apart, both drivers apply brakes and each car retards at 2 m/s^2 . Find the distance between them when both come to rest.

WHY THIS WORKS

Both cars have the same braking model. Find each stopping distance, then compare the combined distance used with the original separation.

METHOD

Each car has $u=20 \text{ m/s}$ and deceleration magnitude 2 m/s^2 . For one car, $s_{\text{stop}}=u^2/(2|a|)=400/4=100 \text{ m}$. Both together consume 200 m of the original 300 m separation. Therefore the distance between them at rest is 100 m.

ANSWER / CHECK

the distance between them at rest is 100 m

CONCEPT TO KEEP

Keep: stopping-distance comparisons are easiest when each body is solved under the same braking model.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q23 EX-M6C-01**HARD**

M-ST06C · Velocity Change + Average Velocity: A Short Route

QUESTION RECAP

During the one-second interval from t to $t+1$, a particle has displacement 125 m and its velocity increases by 50 m/s. Find the distance travelled during the $(t+2)$ th second.

WHY THIS WORKS

A one-second displacement equals the average velocity during that second times one second. With constant acceleration, the velocity change also fixes the acceleration.

METHOD

The interval length is 1 s. Therefore $a=\Delta v/\Delta t=50 \text{ m/s}^2$. Its displacement is 125 m, so average velocity over that interval is 125 m/s. Let the velocity at time t be u ; at $t+1$ it is $u+50$. Then $[u+(u+50)]/2=125$, giving $u=100 \text{ m/s}$ and $v(t+1)=150 \text{ m/s}$. The requested $(t+2)$ th second is the next one-second interval, starting at 150 m/s with $a=50 \text{ m/s}^2$. Its displacement is $150(1)+\frac{1}{2}(50)(1)^2=175 \text{ m}$.

ANSWER / CHECK

Its displacement is $150(1)+\frac{1}{2}(50)(1)^2=175 \text{ m}$

CONCEPT TO KEEP

Keep: over a 1 s interval, displacement equals the interval average velocity; Δv over that second gives a .

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q24 EX-M7B-01**MEDIUM**

M-ST07B · Free Fall from Rest

QUESTION RECAP

A ball is dropped from rest from 18 m. Find the height above ground where $|v|$ is numerically equal to $|g|$, using $g=10 \text{ m/s}^2$.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

$v=10$, $u=0$. From $v^2=2gs$: $100=20s$, so $s=5 \text{ m}$ fallen. Height above ground = $18-5 = 13 \text{ m}$.

ANSWER / CHECK

Height above ground = $18-5 = 13 \text{ m}$

CONCEPT TO KEEP

Keep: in free fall from rest, distance t_2 , so equal distances do not correspond to equal times.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q25 EX-M7B-02**HARD**

M-ST07B · Free Fall from Rest

QUESTION RECAP

A ball is released from height h . Compare the times t_1 and t_2 needed to cover the first and second halves of the distance.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

Let total fall time be T . First half: $t_1=T/2$. Hence $t_2=T-t_1=T(1-1/2)$. Therefore $t_2=(2-1)t_1$ or equivalently $t_1=(2+1)t_2$.

ANSWER / CHECK

$t_2=(2-1)t_1$ or equivalently $t_1=(2+1)t_2$

CONCEPT TO KEEP

Keep: in free fall from rest, distance t_2 , so equal distances do not correspond to equal times.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q26 EX-M7B-03

HARD

M-ST07B · Free Fall from Rest

QUESTION RECAP

Drops fall regularly from 9.8 m. When the first drop hits the floor, the third just starts. Locate the second drop from the floor.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

From $h=9.8$ m and $g=9.8$, first-drop time $T=2$ s. If the third just begins when the first hits, the release gap is $T/2$. The second drop has fallen for $T/2$, so distance fallen $= \frac{1}{2} g(T/2)^2 = 2.45$ m. Height above floor $= 9.8 - 2.45 = 7.35$ m.

ANSWER / CHECK

Height above floor $= 9.8 - 2.45 = 7.35$ m

CONCEPT TO KEEP

Keep: in free fall from rest, distance t^2 , so equal distances do not correspond to equal times.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q27 EX-M7C-01

HARD

M-ST07C · Last-Distance and Between-Point Fall Problems

QUESTION RECAP

A body falling under gravity covers points A and B separated by 80 m in 2 s. Find the distance of A from the starting point, using $g=10$.

WHY THIS WORKS

For freely falling drops, every object follows the same $y \propto t^2$ law but with a different clock start. Translate the spacing into a difference of cumulative falls.

METHOD

$S(t)=5t^2$. Use $5[(T+2)^2 - T^2] = 80 \rightarrow 5(4T+4) = 80 \rightarrow T=3$ s. Therefore distance of A from release $= S(3) = 5 \cdot 9 = 45$ m.

ANSWER / CHECK

distance of A from release $= S(3) = 5 \cdot 9 = 45$ m

CONCEPT TO KEEP

Keep: regularly released drops share the same fall law but have shifted time origins.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q28 EX-M8A-01**EASY****M-ST08A · Highest Point: $v = 0$, $a = -g$** **QUESTION RECAP**

A ball is thrown vertically upward at 150 m/s. With $g=10 \text{ m/s}^2$, the ratio of its velocities after 3 s and 5 s is written as $(x+1)/x$. Find x .

WHY THIS WORKS

For vertical motion with upward positive, velocity changes linearly as $v = u - gt$. Evaluate the velocity at each requested time before forming the ratio.

METHOD

$v_3 = 150 - 10 \cdot 3 = 120 \text{ m/s}$ and $v_5 = 150 - 10 \cdot 5 = 100 \text{ m/s}$. Ratio = $120/100 = 6/5 = (x+1)/x$, giving $x=5$.

ANSWER / CHECK

$x=5$

CONCEPT TO KEEP

Keep: for upward-positive vertical motion, $v = u - gt$.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q29 EX-M8B-01**MEDIUM****M-ST08B · Time to Top, Maximum Height, Flight Time****QUESTION RECAP**

A juggler throws n balls per second and releases each new ball when the previous one reaches its highest point. Find the maximum height.

WHY THIS WORKS

The highest point is identified by $v = 0$. The time to the top and the maximum-height relation connect the release timing to the launch speed and height.

METHOD

Time to top = $1/n$. Thus $u/g = 1/n \rightarrow u = g/n$. Then $H = u^2/(2g) = g/(2n^2)$.

ANSWER / CHECK

Then $H = u^2/(2g) = g/(2n^2)$

CONCEPT TO KEEP

Keep: at the highest point $v = 0$, so time-to-top and maximum height follow directly from constant-gravity relations.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q30 EX-M8B-02

HARD

M-ST08B · Time to Top, Maximum Height, Flight Time

QUESTION RECAP

A ball is thrown upward from a tower at 19.6 m/s and hits the ground after 6 s. Find the greatest height above ground reached by the ball.

WHY THIS WORKS

The highest point is identified by $v = 0$. The time to the top and the maximum-height relation connect the release timing to the launch speed and height.

METHOD

Take upward positive and launch point $y=0$. At ground after 6 s, $y=-h$. So $-h=19.6(6)-4.9(36)=117.6-176.4=-58.8$, hence tower height $h=58.8$ m. Rise above launch point is $u^2/(2g)=19.6$ m. Greatest height above ground = 78.4 m.

ANSWER / CHECK

Greatest height above ground = 78.4 m

CONCEPT TO KEEP

Keep: at the highest point $v = 0$, so time-to-top and maximum height follow directly from constant-gravity relations.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

Q31 EX-M8C-01

HARD

M-ST08C · Same Height Twice

QUESTION RECAP

A ball is thrown vertically upward and reaches maximum height h . Find the ratio of the times at which it is at height $h/3$ while going up and coming down.

WHY THIS WORKS

The same height is reached twice because the upward and downward parts of the trajectory are symmetric in the constant-gravity model. The two roots of the position equation are the two times.

METHOD

At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-\frac{1}{2}gt^2$. Multiply by $6g$: $u^2=6gut-3g^2t^2$. Let $z=gt/u$. Then $3z^2-6z+1=0$, so $z=1\pm6/3$. Therefore $t_{\text{up}}:t_{\text{down}}=(1-6/3):(1+6/3)$, equivalent to $(3-2)/(3+2)$.

ANSWER / CHECK

$t_{\text{up}}:t_{\text{down}}=(1-6/3):(1+6/3)$, equivalent to $(3-2)/(3+2)$

CONCEPT TO KEEP

Keep: a height below the maximum is normally crossed twice - once upward and once downward.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

Q32 EX-M9A-01**HARD**

M-ST09A · Relative Acceleration: Common Gravity Cancels

QUESTION RECAP

A ball is projected upward at 50 m/s at $t=0$. Another is projected upward at 50 m/s at $t=2$ s. Find the meeting time, $g=10$.

WHY THIS WORKS

For delayed launches, use one common global clock or reset the clock explicitly at the second launch. After both are in flight, relative motion can simplify the meeting calculation.

METHOD

At $t=2$ s, Ball 1: $y=50(2)-5(4)=80$ m and $v=50-20=30$ m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and $a_{rel}=0$. Time after second launch $=80/20=4$ s. Global meeting time = 6 s.

ANSWER / CHECK

Global meeting time = 6 s

CONCEPT TO KEEP

Keep: delayed-start problems need a common clock before comparing positions or using relative motion.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q33 EX-M9B-01**MEDIUM**

M-ST09B · Dropped from a Moving Platform

QUESTION RECAP

A gas balloon rises at 10 m/s. At height 75 m a stone is dropped, while the balloon continues upward at the same speed. Find the balloon's height when the stone hits the ground, taking $g=10$.

WHY THIS WORKS

A released object keeps the balloon velocity it had at release. After release, the stone and balloon follow different accelerations and must be tracked with the same time interval.

METHOD

Use upward positive. Stone: $y=75+10t-5t^2$. Setting $y=0$ gives $t=5$ s. During those 5 s, the balloon rises another 50 m. Its height is $75+50=125$ m.

ANSWER / CHECK

Its height is $75+50=125$ m

CONCEPT TO KEEP

Keep: an object released from a moving carrier inherits the carrier velocity at the release instant.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q34 EX-M9C-01

MEDIUM

M-ST09C · Upward vs Downward Launch: Same Impact Speed

QUESTION RECAP

From a building, Ball A is thrown upward and Ball B downward with the same speed magnitude. Compare their velocities on reaching the ground, neglecting air resistance.

WHY THIS WORKS

With no air resistance, final speed at a given height depends on initial speed magnitude and height change, not on whether the initial direction was up or down.

METHOD

Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2 = u^2 + 2gH$. Therefore the impact speed magnitudes are equal. Since both are moving downward at impact, their velocity vectors are also equal in this one-dimensional setup: $v_A = v_B$.

ANSWER / CHECK

Since both are moving downward at impact, their velocity vectors are also equal in this one-dimensional setup: $v_A = v_B$

CONCEPT TO KEEP

Keep: without air resistance, equal initial speed magnitudes from the same height give equal impact speed magnitudes.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

Q35 EX-M9D-02

MEDIUM

M-ST09D · Time Gap: Upward vs Downward Launch from a Tower

QUESTION RECAP

From a tower, upward launch reaches ground in 6 s and equal-speed downward launch in 1.5 s. Find the drop-from-rest time.

WHY THIS WORKS

The upward, downward and dropped cases share the same tower height. Combining the paired launch-time relations removes the launch speed and leaves a relation for the drop time.

METHOD

Use the paired-time relation for the same tower and equal launch-speed magnitude: $t_0 = (t_1 t_2)$. Substituting 6 s and 1.5 s gives $t_0 = (9) = 3$ s.

ANSWER / CHECK

3 s

CONCEPT TO KEEP

Keep: for equal-magnitude upward/downward tower launches, the drop-from-rest time satisfies $t_0 = (t_1 t_2)$.

EXAMSIDE SOURCE

[← RETURN TO QUESTION](#)

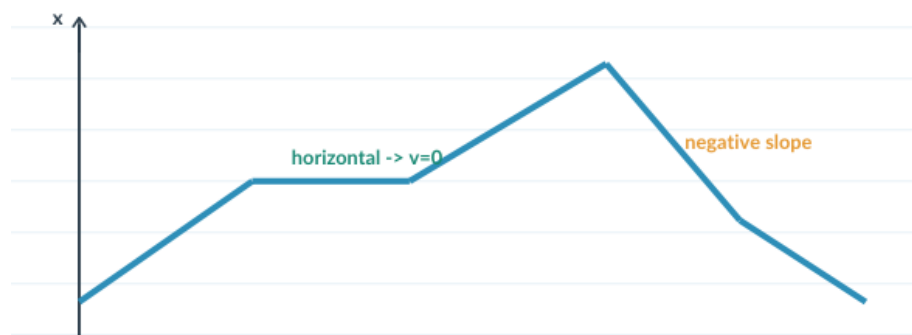
Q36 EX-M10A-01

HARD

M-ST10A · Position-Time Graph: Slope Is Velocity

QUESTION RECAP

From the supplied displacement-time graph, explain how to obtain average velocity and instantaneous velocity using secant and tangent slopes.



WHY THIS WORKS

On an x - t graph, velocity is encoded by slope: a secant slope gives average velocity over an interval, while a tangent slope gives instantaneous velocity.

METHOD

Use endpoint slope for average velocity and tangent slope for instantaneous velocity. The supported set on ExamSIDE is (B), (C), (E).

ANSWER / CHECK

Average velocity is the secant slope between the two times; instantaneous velocity is the tangent slope at the instant. A horizontal tangent means $v = 0$ and a negative slope means negative velocity.

CONCEPT TO KEEP

Keep: x - t slope is velocity; secant = average, tangent = instantaneous.

Q37 EX-M10B-02**MEDIUM**

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

Find distance from $t=0$ to 4 s from a v-t graph.

**WHY THIS WORKS**

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Because velocity is positive over the interval, distance equals the area under the v-t graph. Split the graph into the 0-2 s triangle and the 2-4 s rectangle: $10 \text{ m} + 20 \text{ m} = 30 \text{ m}$.

ANSWER / CHECK

30 m

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

Q38 EX-M10B-03

MEDIUM

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

Find airplane distance in the first 30.5 s from its v-t graph.



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Distance from a v-t graph is obtained by summing the geometric areas under the curve over the requested 30.5 s. Break the graph into the source-supported simple regions, keep the graph units consistent, and add them. The published result is 12 km.

ANSWER / CHECK

12 km

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

Q39 EX-M10B-05

MEDIUM

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

Judge: area under v-t gives distance; area under a-t gives change in velocity.



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Statement I is false in general: signed v-t area is displacement. Statement II is true. Choose 'I false, II true'.

ANSWER / CHECK

Choose 'I false, II true'

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

Q40 EX-M10B-01

HARD

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

From a v-t graph over 40 s, find total distance and average velocity.



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Add absolute triangular areas for distance; add signed areas for displacement. Result: 100 m and average velocity 0.

ANSWER / CHECK

100 m and average velocity 0

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

Q41 EX-M10B-04

HARD

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

Find displacement : distance from a v-t graph over 0-10 s.



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Compute the positive and negative v-t areas separately. Signed area gives displacement = 16 m, while the sum of area magnitudes gives distance = 48 m. Hence displacement : distance = 16:48 = 1:3.

ANSWER / CHECK

1 : 3

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

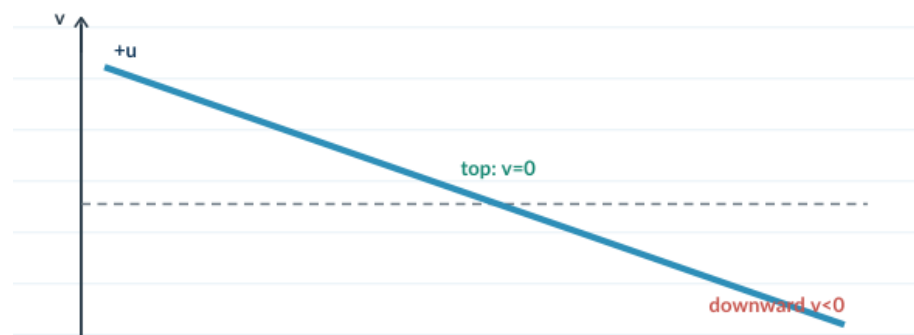
Q42 EX-M10C-01

MEDIUM

M-ST10C · Vertical Projection on a v-t Graph

QUESTION RECAP

Sketch or describe the v-t graph for a body thrown vertically upward.



WHY THIS WORKS

Under constant gravity, velocity changes linearly with time. A collision or inelastic stop appears as a discontinuous jump in velocity, followed by the new motion state.

METHOD

For a vertical throw, acceleration is the constant $-g$. Therefore $v = u - gt$ is a straight line with constant negative slope. It crosses $v = 0$ at the highest point and then continues into negative velocity during the descent.

ANSWER / CHECK

A straight v-t line with constant negative slope; it crosses $v=0$ at the top and continues into negative velocity.

CONCEPT TO KEEP

Keep: constant acceleration gives a straight v-t segment; an instantaneous collision can create a vertical jump.

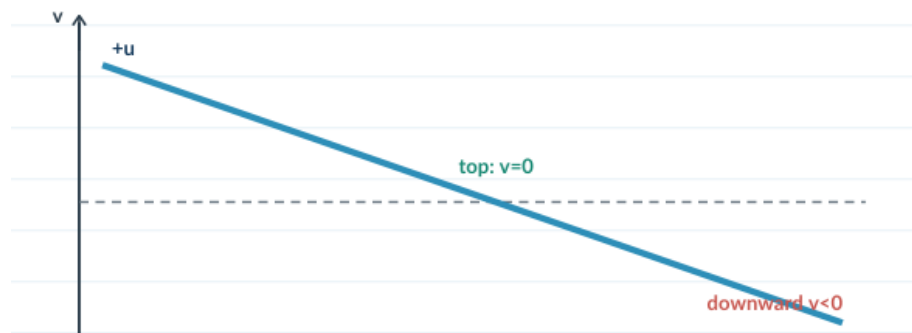
Q43 EX-M10C-02

HARD

M-ST10C · Vertical Projection on a v-t Graph

QUESTION RECAP

Sketch or describe the v-t curve for a bullet moving downward for 10 s, hitting the ground inelastically, then remaining at rest.



WHY THIS WORKS

Under constant gravity, velocity changes linearly with time. A collision or inelastic stop appears as a discontinuous jump in velocity, followed by the new motion state.

METHOD

Before impact, v increases linearly from 100 m/s with slope $+g$. At 10 s the speed is 200 m/s; the inelastic impact makes v jump to 0 and then remain 0.

ANSWER / CHECK

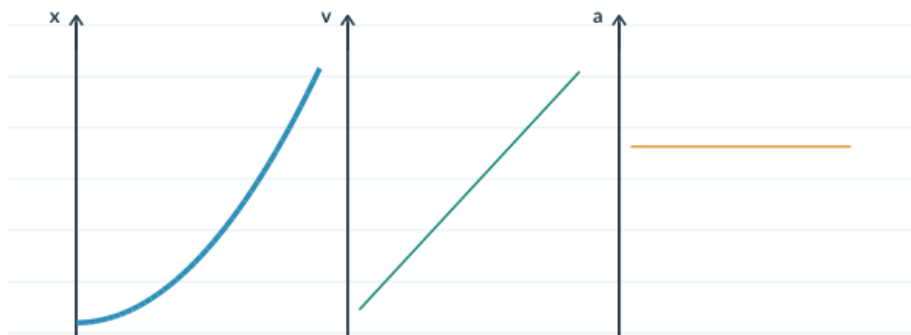
With downward positive: v rises linearly from 100 to 200 m/s by 10 s, drops instantly to 0 at impact, then stays at 0.

CONCEPT TO KEEP

Keep: constant acceleration gives a straight v-t segment; an instantaneous collision can create a vertical jump.

Q44 EX-M10D-01**MEDIUM****M-ST10D · Constant Acceleration: x-t Parabola, v-t Line, a-t Constant****QUESTION RECAP**

Sketch or describe the x-t, v-t and a-t graph shapes for motion with constant acceleration.

**WHY THIS WORKS**

Constant acceleration fixes a family of graph shapes: a-t is constant, v-t is linear and x-t is quadratic. The graphs must tell one consistent motion story.

METHOD

Start from constant acceleration: the a-t graph is horizontal. Integrating the motion story qualitatively, velocity changes linearly with time and position changes quadratically. Therefore x-t is parabolic, v-t is straight and a-t is horizontal.

ANSWER / CHECK

x-t is parabolic, v-t is a straight line, and a-t is horizontal.

CONCEPT TO KEEP

Keep: constant $a \rightarrow$ horizontal a-t, linear v-t, quadratic x-t.

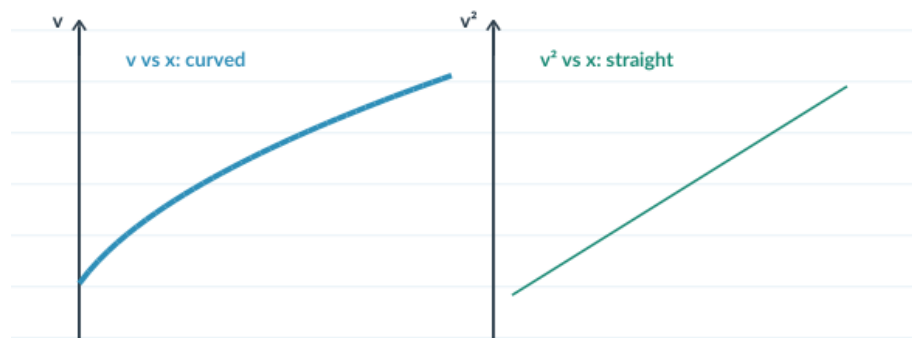
Q45 EX-M10E-03

MEDIUM

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

QUESTION RECAP

A v^2 - x graph is a straight line. Find constant acceleration from its slope.



WHY THIS WORKS

When velocity is given as a function of position, use $a = v \, dv/dx$ to translate a v - x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Compare $v^2 = u^2 + 2ax$: slope of v^2 vs x equals $2a$. Given slope 2, $a = 1 \, \text{m/s}^2$.

ANSWER / CHECK

$1 \, \text{m/s}^2$

CONCEPT TO KEEP

Keep: for $v(x)$, use $a = v \, dv/dx$; do not infer acceleration from the visual slope of v - x alone.

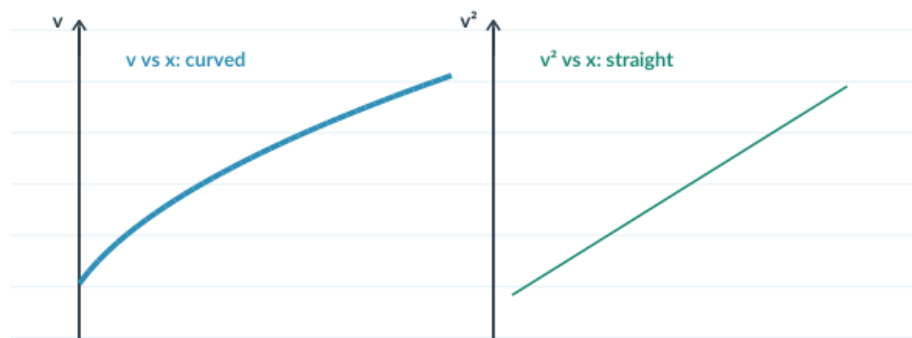
Q46 EX-M10E-04

HARD

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

QUESTION RECAP

For positive velocity with constant negative acceleration, sketch or describe the v - x graph.



WHY THIS WORKS

When velocity is given as a function of position, use $a = v \, dv/dx$ to translate a v - x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Using $v^2 = u^2 + 2ax$ with $a < 0$, v decreases nonlinearly with x . Correct graph is option C.

ANSWER / CHECK

Option C - positive v decreases nonlinearly with x for constant negative acceleration.

CONCEPT TO KEEP

Keep: for $v(x)$, use $a = v \, dv/dx$; do not infer acceleration from the visual slope of v - x alone.

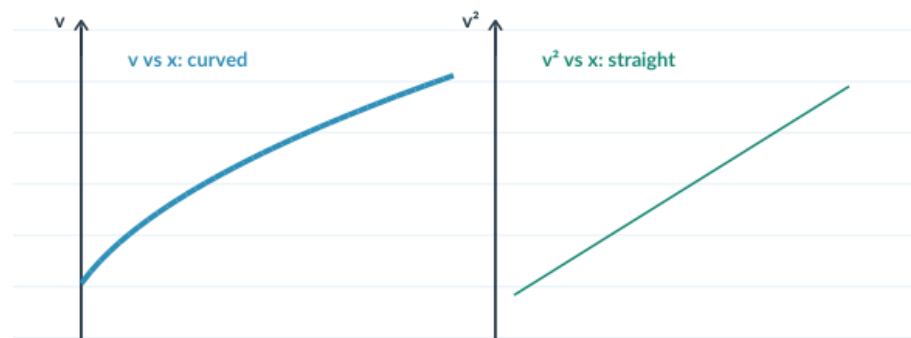
Q47 EX-M10E-01

CHALLENGE

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

QUESTION RECAP

For the given straight-line v - x graph, sketch or describe the corresponding a - x graph.



WHY THIS WORKS

When velocity is given as a function of position, use $a = v \, dv/dx$ to translate a v - x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Use the foundation bridge $a = v \, dv/dx$. For $v = c - mx$, $a = -m(c - mx)$: a straight line with positive slope and negative intercept. ExamSIDE answer: option D.

ANSWER / CHECK

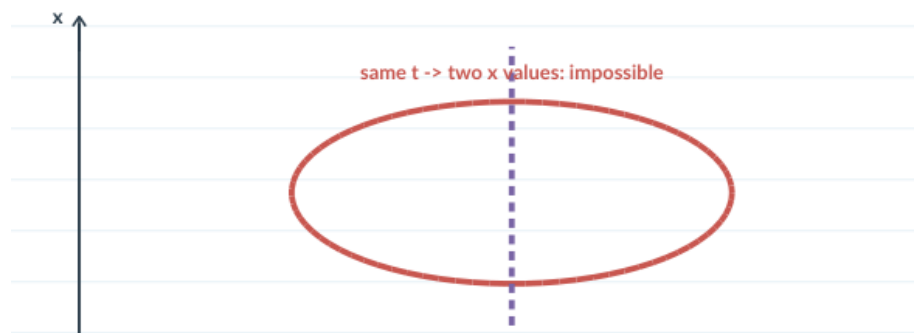
Option D - the a - x graph is a straight line with positive slope and negative intercept.

CONCEPT TO KEEP

Keep: for $v(x)$, use $a = v \, dv/dx$; do not infer acceleration from the visual slope of v - x alone.

Q48 EX-M10F-01**CHALLENGE****M-ST10F · Impossible and Inconsistent Motion Graphs****QUESTION RECAP**

State the single-valued-time test used to decide whether a plotted curve can represent one-dimensional motion. The original option plots are not preserved in the audited source PDF.

**WHY THIS WORKS**

Different motion graphs are valid only if they can describe one consistent one-dimensional state at each instant. Axis meaning, single-valued time and slope relations are the consistency tests.

METHOD

Apply axis meaning and single-valued-time logic. The supported answer is A, B and D only; the excluded curve violates the time/representation constraint.

ANSWER / CHECK

A valid time-based motion graph must be single-valued in time: one particle cannot have two positions or two velocities at the same instant. Apply the vertical-line test and then check consistency with the axis meanings.

CONCEPT TO KEEP

Keep: every graph must respect its axis meaning and describe the same state at the same time.

Q49 EX-M11D-01

HARD

M-ST11D · Ratio of n th-Second Distance to Total Distance

QUESTION RECAP

For motion from rest, the ratio S_{n-1}/S_n of two consecutive one-second distances is written as $1-2/x$ for $n=10$. Find x .

WHY THIS WORKS

For motion from rest with constant acceleration, distances in successive one-second intervals follow the odd-number pattern $1:3:5:\dots$. Ratios therefore depend only on the interval number.

METHOD

For motion from rest, the n th-second distance is proportional to $2n-1$. Thus $s_9:s_{10} = 17:19$. Set $17/19 = 1 - 2/x$; then $2/x = 2/19$, giving $x = 19$.

ANSWER / CHECK

$x = 19$

CONCEPT TO KEEP

Keep: from rest at constant acceleration, successive one-second distances are proportional to $1, 3, 5, \dots, 2n-1$.

Q50 EX-M2B-07

HARD

M-ST02B · Equal Time vs Equal Distance Averages

QUESTION RECAP

A horse rider covers half the distance at 5 m/s. The remaining half is covered at 10 m/s and 15 m/s for equal times. The mean speed is $x/7$ m/s. Find x .

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let total distance be D . First-half time $= D/10$. For the second half, $(10+15)\tau = D/2$, so $\tau = D/50$ and second-half time $= D/25$. Average $= 50/7$ m/s, hence $x=50$.

ANSWER / CHECK

$x=50$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

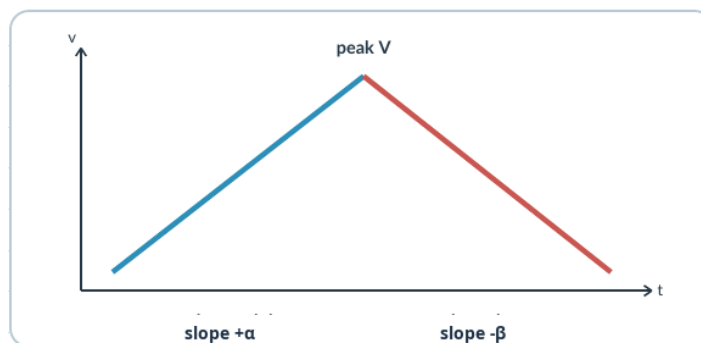
Q51 EX-M4C-02

CHALLENGE

M-ST04C · Average Velocity under Constant Acceleration

QUESTION RECAP

A car accelerates from rest at constant rate α for some time, then decelerates at constant rate β to come to rest. If total time is T , express the total distance in terms of α , β and T .



WHY THIS WORKS

The two stages meet at one common peak velocity. In a v - t picture the whole journey is one triangle, so time relations determine the peak and triangle area gives distance.

METHOD

Let V be the peak velocity. During acceleration, $t_1 = V/\alpha$. During braking, $t_2 = V/\beta$. Thus $T = V(1/\alpha + 1/\beta)$, so $V = \alpha \beta T / (\alpha + \beta)$. The complete v - t graph is a triangle, therefore $S = \frac{1}{2} V T$. Hence $S = \alpha \beta T^2 / [2(\alpha + \beta)]$.

ANSWER / CHECK

$$S = \alpha \beta T^2 / [2(\alpha + \beta)]$$

CONCEPT TO KEEP

Keep: a linear accelerate-then-brake v - t graph forms a triangle; its area is total displacement.

Q52 EX-M4E-04

HARD

M-ST04E · Read a Motion Equation from Its Coefficients

QUESTION RECAP

Two buses start together with positions $x_P = \alpha t + \beta t^2$ and $x_Q = f t - t^2$. Find when their velocities are equal.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + \frac{1}{2}at^2$. This reveals u and a without using calculus.

METHOD

Read $u_P = \alpha$, $a_P = 2\beta$; $u_Q = f$, $a_Q = -2$. Set $\alpha + 2\beta t = f - 2t$, so $t = (f - \alpha) / [2(\beta + 1)]$.

ANSWER / CHECK

$$t = (f - \alpha) / [2(\beta + 1)]$$

CONCEPT TO KEEP

Keep: match a quadratic $x(t)$ to $x_0 + ut + \frac{1}{2}at^2$ to read u and a directly.

Q53 EX-M9A-02

CHALLENGE

M-ST09A · Relative Acceleration: Common Gravity Cancels

QUESTION RECAP

Two equal spherical balls are thrown upward 3 s apart with the same initial velocity 35 m/s. Each radius is 5 cm. Find the collision height, $g=10$.

WHY THIS WORKS

For delayed launches, use one common global clock or reset the clock explicitly at the second launch. After both are in flight, relative motion can simplify the meeting calculation.

METHOD

At the second launch (3 s), Ball 1 is at 60 m and moving upward at 5 m/s. Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant. Allowing for the 10 cm centre separation at contact gives a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m.

ANSWER / CHECK

a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m

CONCEPT TO KEEP

Keep: delayed-start problems need a common clock before comparing positions or using relative motion.

Q54 EX-M10B-06

HARD

M-ST10B · Velocity-Time Graph: Slope, Area, Distance

QUESTION RECAP

From a v-t graph, find distance/displacement ratio in 25 s.



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Use signed v-t area for displacement and absolute area for distance. The source graph gives distance = 250 m and displacement = 150 m, so distance/displacement = $250/150 = 5/3$.

ANSWER / CHECK

5/3

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

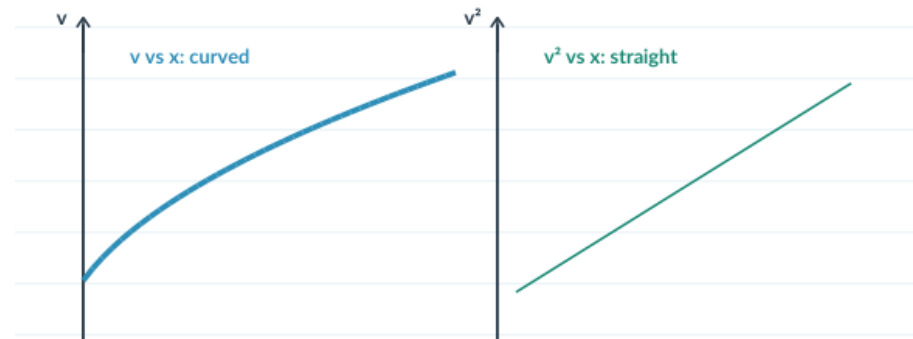
Q55 EX-M10E-02

CHALLENGE

M-ST10E · Velocity vs Displacement: Read the Axes Before Naming the Curve

QUESTION RECAP

For a linear v - x graph, sketch or describe the corresponding a - x graph.



WHY THIS WORKS

When velocity is given as a function of position, use $a = v \, dv/dx$ to translate a v - x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Use $a = v \, dv/dx$. Result is a straight line with positive slope and negative intercept; ExamSIDE explanation selects option A.

ANSWER / CHECK

Option A - the a - x graph is a straight line with positive slope and negative intercept.

CONCEPT TO KEEP

Keep: for $v(x)$, use $a = v \, dv/dx$; do not infer acceleration from the visual slope of v - x alone.

Q56 EX-M4B-03

HARD

M-ST04B · Displacement under Constant Acceleration

QUESTION RECAP

A body starts from rest with constant acceleration. It covers S_1 in the first $(p-1)$ s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

WHY THIS WORKS

From rest under constant acceleration, cumulative displacement is proportional to t^2 . That lets two cumulative distances be combined without first solving for the acceleration.

METHOD

Let cumulative displacement from rest be Kt^2 , where $K = \frac{1}{2}a$. Then $S_1 = K(p-1)^2$ and $S_2 = Kp^2$. If $S_1 + S_2$ is covered in time T , $KT^2 = K[(p-1)^2 + p^2]$. Thus $T^2 = 2p^2 - 2p + 1$, so $T = (2p^2 - 2p + 1)$ s.

ANSWER / CHECK

$T = (2p^2 - 2p + 1)$ s

CONCEPT TO KEEP

Keep: from rest at constant acceleration, $S(t) \propto t^2$.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q57 EX-M4D-04

HARD

M-ST04D · Remove Time: Why Velocities Become Squared

QUESTION RECAP

A train accelerates uniformly. Its engine passes a signal with speed u , and the last carriage passes the same signal with speed v . What is the speed w when the midpoint of the train passes the signal?

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2 .

METHOD

Let train length be L . When the engine passes the signal, speed is u . When the last carriage passes, the engine has advanced L , so $v^2 = u^2 + 2aL$. At the midpoint passage, displacement is $L/2$, so $w^2 = u^2 + 2a(L/2) = u^2 + aL$. From the full-length equation, $aL = (v^2 - u^2)/2$. Therefore $w^2 = (u^2 + v^2)/2$ and $w = [(u^2 + v^2)/2]$.

ANSWER / CHECK

$w^2 = (u^2 + v^2)/2$ and $w = [(u^2 + v^2)/2]$

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EXAMSIDE SOURCE

<- RETURN TO QUESTION

Q58 EX-M7C-02

CHALLENGE

M-ST07C · Last-Distance and Between-Point Fall Problems

QUESTION RECAP

Water droplets fall regularly. The spacing between a droplet observed at its 4th second of fall and the next droplet is 34.3 m. Determine the drop rate, using $g=9.8$.

WHY THIS WORKS

For freely falling drops, every object follows the same $y \propto t^2$ law but with a different clock start. Translate the spacing into a difference of cumulative falls.

METHOD

Let release interval be τ . The older drop has fallen 4 s; the next one has fallen $4-\tau$ s. Spacing = $4.9[16-(4-\tau)^2]=34.3$. Divide by 4.9: $16-(4-\tau)^2=7$, so $(4-\tau)^2=9$ and $\tau=1$ s. Therefore the rate is 1 drop/s.

ANSWER / CHECK

the rate is 1 drop/s

CONCEPT TO KEEP

Keep: regularly released drops share the same fall law but have shifted time origins.

Q59 EX-M9D-01

HARD

M-ST09D · Time Gap: Upward vs Downward Launch from a Tower

QUESTION RECAP

A body thrown upward from a tower reaches ground in t_1 . Thrown downward with the same speed it reaches in t_2 . Find the time if simply dropped.

WHY THIS WORKS

The upward, downward and dropped cases share the same tower height. Combining the paired launch-time relations removes the launch speed and leaves a relation for the drop time.

METHOD

The upward and downward launches use the same tower height and the same launch-speed magnitude. Combining the two constant-gravity time relations eliminates the launch speed. The source-supported paired-time result is $t_0^2 = t_1 t_2$, so $t_0 = (t_1 t_2)$.

ANSWER / CHECK

$$t_0 = (t_1 t_2)$$

CONCEPT TO KEEP

Keep: for equal-magnitude upward/downward tower launches, the drop-from-rest time satisfies $t_0 = (t_1 t_2)$.

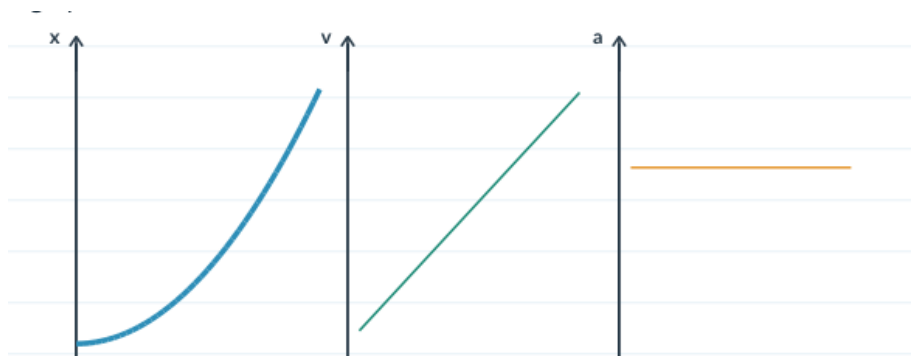
Q60 EX-M10D-02

HARD

M-ST10D • Constant Acceleration: x-t Parabola, v-t Line, a-t Constant

QUESTION RECAP

From rest with uniform positive acceleration, identify all correct qualitative graphs.



WHY THIS WORKS

Constant acceleration fixes a family of graph shapes: a-t is constant, v-t is linear and x-t is quadratic. The graphs must tell one consistent motion story.

METHOD

From rest with constant positive acceleration, a is a positive constant, so a-t is horizontal above zero. Then $v = at$ is linear through the origin and $x = \frac{1}{2}at^2$ is an upward-opening parabola. Compare each candidate graph with these three requirements.

ANSWER / CHECK

For motion from rest with uniform positive acceleration: x-t is an upward-opening parabola through the origin, v-t is a straight increasing line through the origin, and a-t is a horizontal line above zero. (The source labels the supported candidates A, B and D.)

CONCEPT TO KEEP

Keep: constant $a \rightarrow$ horizontal a-t, linear v-t, quadratic x-t.

Q61 EX-M10F-02**CHALLENGE****M-ST10F · Impossible and Inconsistent Motion Graphs****QUESTION RECAP**

State the consistency test for multiple graph representations of the same motion. The audited source records graph 3 as the inconsistent source option.

**WHY THIS WORKS**

Different motion graphs are valid only if they can describe one consistent one-dimensional state at each instant. Axis meaning, single-valued time and slope relations are the consistency tests.

METHOD

Translate each graph into the same motion story. Official-solution sources identify graph 3 as inconsistent because its initial slope implies zero initial velocity while the others do not.

ANSWER / CHECK

The source records graph 3 as inconsistent because its initial slope implies zero initial velocity while the other representations describe a nonzero initial velocity. The reusable test is to translate every graph into one common motion story and compare initial state, signs, turning points and timing.

CONCEPT TO KEEP

Keep: every graph must respect its axis meaning and describe the same state at the same time.